

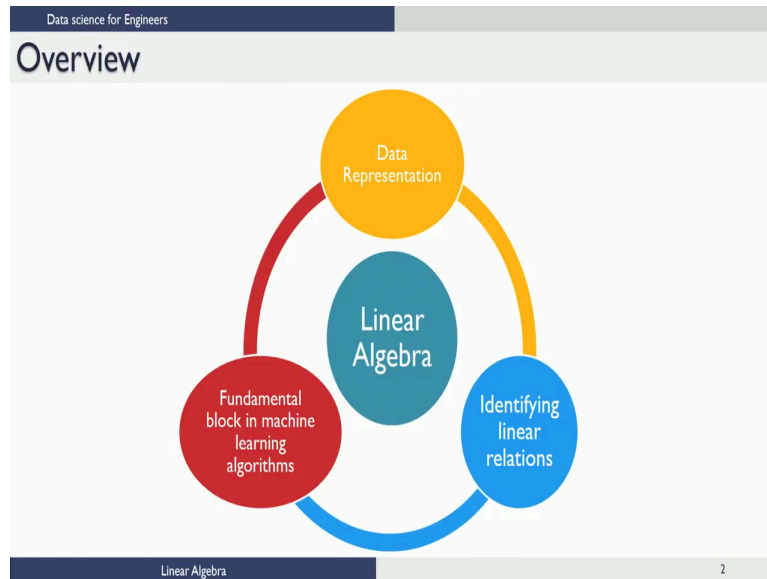
Data Science for Engineers
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Lecture – 12
Linear Algebra for Data science

This lecture is on linear algebra for data science. Linear algebra is a very fundamental part of data science and usually a typical one semester linear algebra course will run for about 36 hours. What we are trying to do here is we are trying to introduce the use of linear algebra in data science in a few hours. So, that necessarily means that we cannot cover the topic of linear algebra in all its detail; however, what we have attempted to do here is to identify the most important concepts from linear algebra, that are useful in the field of data science and in particular for the material that we are going to teach in this course. So, in that sense we have crystallized a few important concepts from linear algebra that the participants should learn and understand.

So, that is one thing that I would like to mention right at the beginning. The second thing that I would like to mention is the following; Linear algebra can be treated very theoretically very formally; however, in the short module on linear algebra which has relevance to data science, what we have done is we have tried to explain the ideas in as simple fashion as possible without being too formal. However, we do not do any hand waving we teach linear algebra in a simple fashion. So, that is another thing that I would like you to remember as we go through this material. So, we first start by explaining what linear algebra is useful for.

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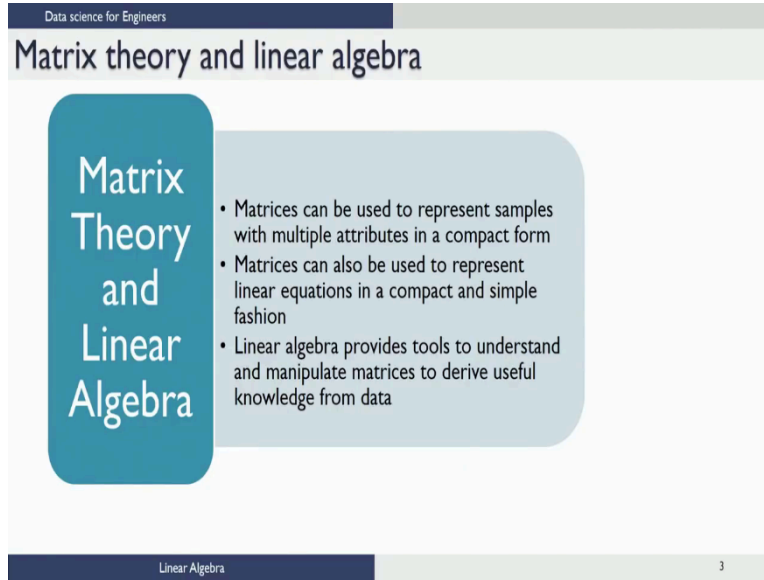


So, when one talks about data science, Data Representation becomes an important aspect of data science and data is represented usually in a matrix form and we are going to talk about this representation and concepts in matrices. The second important thing that one is interested from a data science perspective is, if this data contains several variables of interest, I would like to know how many of these variables are really important and if there are relationships between these variables and if there are these relationships, how does one un-cover these relationships?

So, that is also another interesting and important question that we need to answer from the viewpoint of understanding data. Linear algebraic tools allow us to understand this and that is something that we will teach in this course. The third block that we have basically says that the ideas from linear algebra become very very important in all kinds of machine learning algorithms.

So, one needs to have a good understanding of some of these concepts be-fore you can go and understand more complicated or more complex machine learning algorithms. So, in that sense also linear algebra is an important component of data science. So, we will start with matrices. Many of you would have seen matrices before. I am going to look at matrices and summarize the most important ideas that are relevant from a data science viewpoint. What is a matrix? Matrix is a form of organizing data into rows and columns. Now, there are many ways in which you can organize data. A matrix provides you a convenient way of organizing this data.

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Matrix theory and linear algebra

Matrix Theory and Linear Algebra

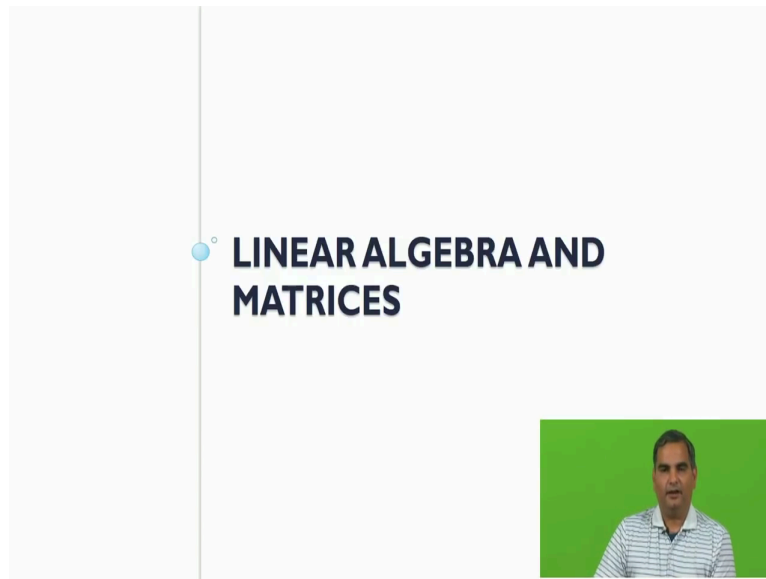
- Matrices can be used to represent samples with multiple attributes in a compact form
- Matrices can also be used to represent linear equations in a compact and simple fashion
- Linear algebra provides tools to understand and manipulate matrices to derive useful knowledge from data

Linear Algebra 3

So, if you are an engineer and you are looking at data for multiple variables, at multiple times, how do you put this data together in a format that can be used later, is what a matrix is helpful for.

Now, matrices can be used to represent the data or in some cases matrices can also be used to represent equations and the matrix could have the coefficients in several equations as its component. Now, once we generate these matrices then you could use the linear algebra tools to understand and manipulate these matrices, so that you are able to derive useful information and useful knowledge from this data.

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So, let us start and then try and understand how we can understand and study matrices.

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Matrices for data science: Data representation

- Usually matrices are used to store and represent the data on machines
- Matrix is a very natural approach for organizing data
- In general, data is organized in the following fashion
 - Rows represent samples
 - Columns represent the values of the variables (or attributes)
 - It is also possible to use rows for variables and columns for samples
 - However, we will stick to rows as samples and columns as variables in all of the material that will be presented

Linear Algebra
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As I mentioned before matrices are usually used to store and represent data on machines and matrix is a very natural approach for organizing data. Typically when we have a matrix it is a rectangular structure with rows and columns. In general, we use the rows to represent samples and I will explain what I mean by this in subsequent slides and we use columns to represent the variables or attributes in the data.

Now, this is just one representation. It is possible that you might want to use rows to represent variables and columns to represent samples and there is nothing wrong with that; however, in this course and all the material that we present in this course we will stick to using

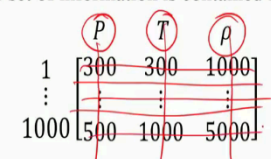
rows to represent samples and columns to represent variables as far as this course is concerned.

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Data representation: Examples

- A real life example
 - Consider a reactor which needs to be controlled using multiple attributes from various sensors like Pressure (Pa), Temperature (K), Density (gm/m^3) etc.
 - Independently, the sensors have generated 1,000 data points
 - This complete set of information is contained in





Linear Algebra

Let me explain matrix using a real life example. Let us consider that you are an engineer and you are looking at a reactor which has multiple attributes and you are getting information from sensors such as pressure sensors, temperature sensors and density and so on.

Now let us assume that you have taken 1000 samples of these variables. Now you want to organize this data somehow. So that you can use it for purposes needed. One way to do this is to organize this in this matrix form, where the first column is the column that corresponds to the values of pressure at different sample points. The second column corresponds to the value of temperature at several sample points and the third column corresponds to the value of density at several sample points.

So, that is what I meant when I said the columns are used to represent the variable. So, each column represents a variable column 1 pressure, column 2 temperature and column 3 density and when you look at the rows; the first row represents the first sample.

Here in the first sample you will read that the value of pressure was 300, the value of temperature was 300 and the value of density was 1000. Similar to that you will have many rows corresponding to each sample point up to the last row 1000th row, which is a 1000 sample point; which has a pressure is 500 temperature is 1000 and density is 5000.

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Data representation: Examples

- Example 2:

$$X = [1, 2, 3]^T$$

$$Y = [2, 4, 6]^T$$
- X and Y are vectors pertaining to some attributes
- We define the A matrix using a column bind of X and Y thus representing data in a matrix format (the code for the same is attached)

R Code

```
x=c(1,2,3)
y=c(2,4,6)
A=cbind(x,y)
print(A)
```

1	2
2	4
3	6

Output

```
print(A)
  x y
[1,] 1 2
[2,] 2 4
[3,] 3 6
```



Linear Algebra

Let us take another example let us say I have 2 vectors, $X = [1, 2, 3]^T$ and $Y = [2, 4, 6]$. Let us say this is some variable that you have measured and Y is some other variable you have measured and the 3 values could represent the 3 sampling points at which you measured these.

Now, in R if you want to put this numbers into a matrix, it is a very very simple code. What you do is: $X = c(1, 2, 3)$ it tells you it is a column vector with values 1, 2, 3 y is a column vector with values 2, 4, 6 and then you use the command `A = cbind(x, y)` which puts these together and when you print A you get the value of this matrix.

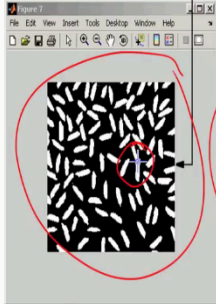
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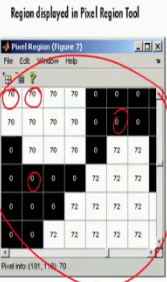
Data representation: Examples

- The simplicity in representation will become apparent when the image below is considered

Pixel Region rectangle



Region displayed in Pixel Region Tool





Linear Algebra

Now, we have been talking about using matrices to represent data from engineering processes sensors and so on. The notion of matrix and manipulating matrices is very important for all kinds of applications. Here is another example where I am showing how a computer might store data about pictures. So, for example, if you take this picture here on this left hand side and you want to represent this picture somehow in a computer. One way to do that would be to represent this picture as a matrix of numbers.

So, in this case for example, if you take a small region here you can break this small region into multiple pixels and depending on whether a pixel is a white background or a black background you can put in a number. So, for example, here you see these numbers which are large numbers which represent white back-ground and you have these small numbers which represent black background. So, this would be a snapshot or a very small part of this picture.

Now, when you make this picture into many such parts you will have a much larger matrix and that larger matrix will start representing the picture. Now, you might ask; why would I do something like that? There are many many applications where you want the computer to be able to look at different pictures and then see whether they are different or the same or identify sub components in the picture and so on. And all of those are done through some form of matrix manipulation and this is how you convert the picture into a matrix.

Now notice that while we converted this matrix we have  into a rectangular form, where we have rows and columns, where data is filled as a representation for this picture.

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Data representation: Examples

- Storing
 - The image is stored in the machine as a large matrix of pixel values across the image.
 - Thus, storing the pixel value matrix is equivalent to storing the image for the machine
- Identification
 - Several machine learning algorithms are deployed in order to “teach” the machine how to identify a particular image.
 - Linear algebra and matrix operations are at the heart of these machine learning algorithms.

Linear Algebra 9

So, the image that I showed before could be stored in the machine as a large matrix of pixel values across the image. And you could show other pictures and then say are these pictures similar to this, are these dissimilar, how similar or dissimilar and so on and the ideas from linear algebra matrix operations are at the heart of these machine learning algorithms.

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Data as matrix: Summary

```

graph LR
    DM[Data Matrix] --- R[Rows]
    DM --- C[Columns]
    R --- S[Samples Data points]
    C --- A[Attributes Variables]
  
```

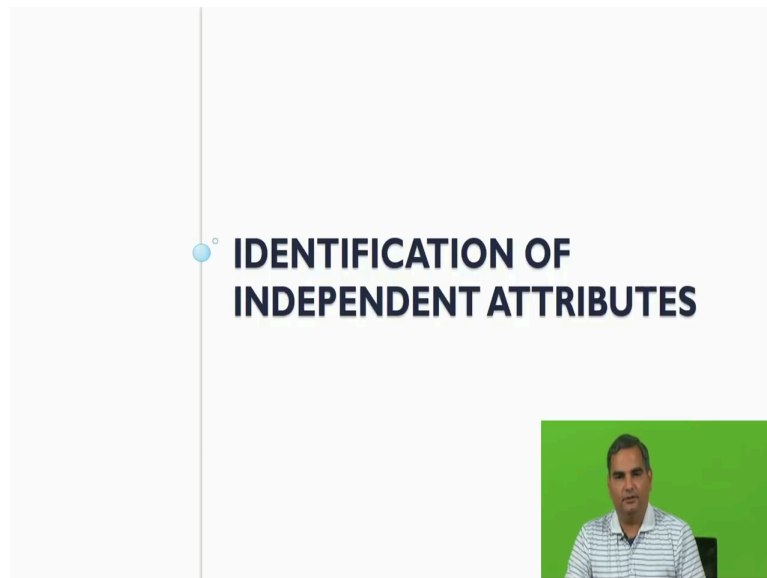
The diagram illustrates the structure of a data matrix. A central blue box labeled 'Data Matrix' is connected by yellow lines to two orange boxes: 'Rows' and 'Columns'. The 'Rows' box is further connected by a red line to a red box labeled 'Samples (Data points)'. The 'Columns' box is connected by a red line to a red box labeled 'Attributes (Variables)'. Both red boxes have a red oval around them. In the bottom right corner, there is a small video inset showing a man speaking.

Linear Algebra

So, in summary if you have a data matrix- the data matrix could be data from census in an engineering plan. It could be data which represents a picture; it could be data which is representing the model where you have the coefficients from several equations. So, the matrix

basically could have data from various different sources or various different viewpoints and each data matrix is characterized by rows and columns and the rows represent samples and the columns represents attributes or variables.

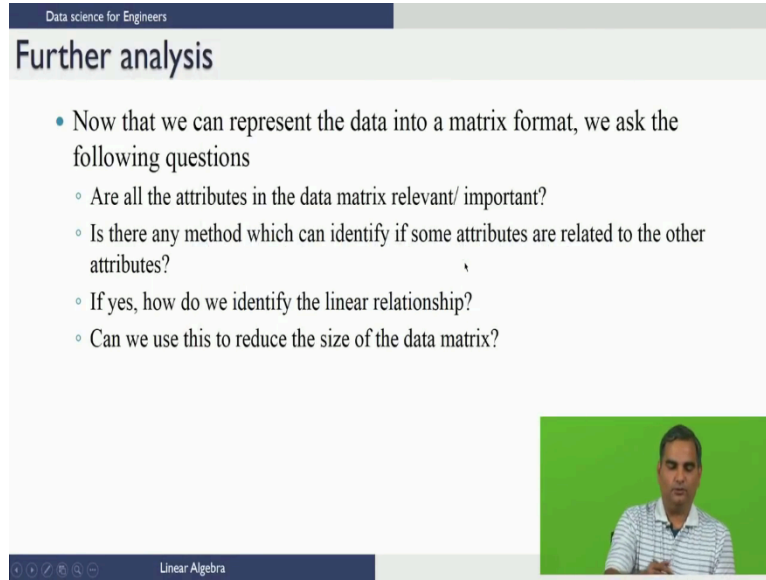
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Now that we have understood how we generate a matrix and why we generate matrices. The next question might be the following, supposing I have a matrix where I have several samples and several variables, I might be interested in knowing if all the variables that are there in the data are important. In other words, I would really like to know of all these variables, how many are actually independent variables?

So that I know how much information is there in this data. So, if let us say I have thousands of variables and of those there are only 4 or 5 that are independent. Then it means that actually I can store values for only these few variables and calculate the remaining as a function of these variables. So, it is important to know how much information I actually have.

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Further analysis

- Now that we can represent the data into a matrix format, we ask the following questions
 - Are all the attributes in the data matrix relevant/ important?
 - Is there any method which can identify if some attributes are related to the other attributes?
 - If yes, how do we identify the linear relationship?
 - Can we use this to reduce the size of the data matrix?

Linear Algebra

So, this would lead to the following questions. The first question might be; Are all the attributes or variables in the data matrix really relevant or important? Now a sub question is to say are they related to each other. If I can write one variable as a combination of other variables then basically I can drop that variable and retain the other variables and calculate this variable whenever I want.

So, that is a very important idea that we want to use in machine learning and various other applications. So, how do I find out how many of these variables are really independent and let us assume that I do and that only a few variables are really independent, then how do I identify the relationship between these variables and the other dependent variables and once I do that how do we actually reduce the size of the data matrix and so on; are questions that one might be interested in answering.

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Identification of independent attributes: Example

- Consider the ideal reactor example with multiple (say, 4) attributes like Pressure, Temperature, Density, Viscosity, etc. with 500 samples.
- Thus we have a 500×4 matrix such that

$$A = [P \ T \ D \ \eta]$$
- P, T, D and η are vectors of 500 samples from the pressure, temperature, density and viscosity sensors.
- How does one identify the number of independent attributes?

Linear Algebra 13

So, let us consider the example that we talked about; the reactor with multiple attributes. In the previous slide, we talked about pressure, temperature and density. Here I have also included viscosity. Let us say I have 500 samples. Then when I organize this data with the variables in the columns and samples in the row, then I will get a 500 by 4 matrix, where each row represents one of the 500 samples and if you go across the column, it will represent the variable values, at all the samples that we have taken. Now, I want to know how many of these are really independent attributes.

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Identification of independent attributes: Example

- Domain knowledge

$$D \sim f(P, T)$$
- Thus, in some sense D is a function of P and T
- Implying that at least one attribute is dependent on the others
- This variable can be calculated as a linear combination of the other variables
- The physics of the problem helps us identify the relationship in the data matrix
- We now ask if the data itself will help us identify these relationships

Linear Algebra 14

So, from domain knowledge it might be possible to say that density is in general a function of pressure and temperature. So, this implies that at least one attribute is dependent on the other and if this relationship happens to be a linear relationship then this variable can be calculated as a linear combination of the other variables.

Now, if all of this is true then the physics of the problem has helped us identify the relationship in this data matrix. The real question that we are interested in asking is if the data itself can help us identify these relationships.

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Number of independent attributes: Rank of a matrix

- Let us assume that we have many more samples than attributes for now
- Is there any approach which can be used to identify the number of linear relationships between the attributes purely using data?
- This is addressed by the concept of the **rank** of the matrix.
- **Rank** of a matrix refers to the number of linearly independent rows or columns of the matrix
- The rank of a matrix can be found using the rank command: `rank(A)`

Linear Algebra
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Let us first assume that we have many more samples than attributes for now and once we have the matrix, when we want to identify the number of independent variables. The concept that is useful is the rank of the matrix and the rank of the matrix is defined as the number of linearly independent rows or columns that exist in the matrix.

And once you identify these number of linearly independent rows or columns then you could basically say that I have only so many independent variables and the remaining are dependent variables and the rank of the matrix can be easily found using the rank command in our rank of A.

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Rank of a matrix: Example 2

- Consider another example

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 0 \\ 3 & 6 & 0 \end{pmatrix}$$

We observe that

- (Col. 2) = 2 x (Col. 1)
- (Col. 3) is independent

Thus, the rank of this matrix is 2

R Code

```
A=matrix(c(1,2,3,2,4,6,1,0,0),ncol=3,byrow=F)
library(pracma)
Rank(A)
```

Output

```
> Rank(A)
[1] 2
```

Linear Algebra



So, consider this example here where I have this a matrix which is 1, 2, 3, 2, 4, 6, 1, 0, 0. If you notice this matrix has been deliberately generated such that the second column is twice column one.

So, in other words the second column is dependent on the first column or you could say the first column is dependent on the second column. Now, there is one other column which is independent of these two so, if you think about this matrix there are 2 independent columns; which basically means there are 2 independent variables. So, if you were to use R to identify this, simply load the correct library and then use the command rank of A and you will get the rank of the matrix to be 2.

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Rank: Advantages and summary

Rank

- Work with reduced set of variables
- Dependent attributes can be calculated if they are from the same data generation process
- Independent of the size of the dataset if the data is from the same data generation process

Linear Algebra

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So, the notion of rank is important. It allows us to work with a reduced set of variables and once we identify the independent variables, the dependent variables or attributes can be calculated from the independent variables, if the data is being generated from the same data generation process.

And if you identify that there are certain variables which are dependent on other variables and as long as the data generation process is the same, it does not matter how many samples that you generate, you can always find the de-pendent variables as a function of the independent variables.

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Now that we have talked about identifying how many independent variables are there; assume that the number of independent variables are less than the number of variables. Then that basically automatically means that there are really linear relationships between these variables.

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Linear relationships among attributes

- Now that we have identified the number of linearly independent attributes:
 - How does one identify those linear relations among the attributes?
- Such questions are addressed by the linear algebraic concepts of null space and nullity



Linear Algebra

Now we ask the next question as to how do we identify these linear relationships among variables or attributes. So, this question of how does one identify the linear relationships among attributes, is answered by the concepts of null space and nullity which is what we going to describe now.

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Null space for data science

- The null space of a matrix A consists of all vectors β such that $A\beta = 0$ and $\beta \neq 0$
- Nullity of a matrix is the number of vectors in the null space of the given matrix
- The size of the null space of a matrix provides us with the number of linear relations among the attributes
- And the null space vectors β are useful to identify these linear relationships

$$\begin{matrix} A & \beta & = & (3 \times n) & 5 \\ 3 \times 3 & 3n & & \downarrow 0 & \textcircled{2} \end{matrix}$$



Linear Algebra

When we have a matrix A and if we are able to find vectors β such that $A\beta = 0$ and $\beta \neq 0$ then we would call this vector β as being the null space of the matrix. So, let us do some simple numbers here for example, if A is a 3 by 3 matrix because β multiplies A , β has to be 3 by 1 and the resultant will be some 3 by 1 vector and if all the elements of this 3 by 1 vector are 0, then we would call this β as being the null space of the matrix. Now interestingly the size of the null space of the matrix provides us with the number of relationships that are among the variables.

If you have a matrix which is of dimension 5 and let us say the size of null space is 2, then this basically means that there are 2 relationships among these 5 variables, which also automatically means that of these 5 variables only 3 are linearly independent because the 2 relationships would let you calculate the dependent variables as a function of these independent variables.

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Data science for Engineers

Null space : general description

- Let us suppose
- $A = \begin{bmatrix} x_{11} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots \\ x_{m1} & \cdots & x_{mn} \end{bmatrix}$ is a data matrix and there is one vector in the null space of A , i.e., $\beta = [\beta_1 \ \dots \ \beta_m]^T$, then as per the definition, β satisfies all the equations given below

$x_{11}\beta_1 + x_{12}\beta_2 + \dots + x_{1n}\beta_n = 0$

- $x_{11}\beta_1 + x_{12}\beta_2 + \dots + x_{1n}\beta_n = 0$
- :
- $x_{m1}\beta_1 + x_{m2}\beta_2 + \dots + x_{mn}\beta_n = 0$



Now, let us look at this in little more detail to understand how we can use this null space vectors. Let us assume that I have a matrix such as this. Now if there is a β which is what we have written here such that $A\beta = 0$. As I mentioned in the previous slide, if I have this matrix of this dimension and if I multiply β with this matrix, then on the right hand side there are going to be several elements and for β to be the null space of this matrix every one of the elements has to be equal to 0. Now, let us look at what each of these elements are equal to.

So, if you take the first element on the right hand side, that would be a product of the first row of this data matrix and this β vector which would basically be $x_{11} \beta_1 + x_{12} \beta_2$; all the way up to x_n and $\beta_n = 0$. Now, similarly if you get to the second row and multiply the second row by this vector you will get another equation.

So, if we keep going down, for every sample if you write this product, you are going to get an equation. The last sample for example, will be $x_{m1} \beta_1 + x_{m2} \beta_2 + \dots + x_{mn} \beta_n = 0$. Now, there is something interesting that you should notice here. This equation seems to be satisfied for all samples. So, what this basically means is, irrespective of the sample, the variables seem to hold this equation and since this equation is held for every sample we would assume that this is a true relationship between all of these variables or attribute. So, in other words this β_1 to β_m gives you in some sense a model equation or a relationship among these variables.

So, one might say that this equation can generally be written as $x_1 \beta_1 + x_2 \beta_2 + \dots + x_n \beta_n = 0$; where you can take any sample and substitute the values of the variables in that sample at x_1, x_2 up to x_n and this is to be satisfied. So, this is a true relationship.

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Null space: The idea

- Notice that if $A\beta = 0$, every row of A when multiplied by β goes to zero
- This implies that variable values in each sample (represented by a row) behave the same
- This helps in identifying the linear relationships in the attributes
- Every null space vector corresponds to one linear relationship
- This idea is demonstrated further using examples

Linear Algebra 22

So, this is what we have said. Again here notice that if $A\beta = 0$ every row of A when multiplied by β goes to 0.

So, this implies that the variable values in each sample behave the same so, we have truly identified a linear relationship between these variables. Now, every null space vector corresponds to one such relationship and if you have more vectors in the null space then you have more relationships that you can uncover.

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Rank nullity theorem

- Consider the data matrix A with the null space and nullity as defined before
- The rank- nullity theorem helps us to relate the nullity of the data matrix to the rank and the number of attributes in the data
- According to the rank-nullity theorem

Nullity of A + Rank of A = Total number of attributes of A

Number of equations + Number of independent variables = Total number of variables

Linear Algebra

So, we will demonstrate this example this with a further example. So, this rank nullity theorem basically says the nullity of matrix A + the rank of matrix A is going to be equal to the total number of attributes of A or the number of columns of the matrix. So, the nullity of a tells you how many equations are there; there are so, many vectors in the null space.

The rank of A tells you how many independent variables are there and when you add these two you should get the total number of variables that is there in your problem.

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Summary till now

- Data Matrix**
 - The available data is expressed in the form of a data matrix
 - This data matrix is further used to do the necessary operations
- Null Space**
 - Defined as a collection of vectors satisfying $A\beta = 0$
 - Helps in identifying the linear relationships between the attributes directly
- Nullity**
 - Nullity is the size of the null space of the data matrix
 - Useful to identify the number of linear relationships in the attributes
 - Rank- Nullity theorem

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So, to summarize, when you have data, the available data can be expressed in the form of a data matrix and as we saw in this lecture this data matrix can be further used to do different types of operations. We also defined null space, null space is defined as a collection of vectors that satisfy this relationship $A \text{ times } \beta = 0$.

So, this basically helps in identifying the linear relationships between the attributes or the variables directly and the number of such vectors or number of such relationships is what is given by the nullity. The Nullity of the matrix tells you how many relationships are there or how many vectors are there in the null space.

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Null space: An Example

- Consider the matrix A with attributes $\{x_1, x_2\}$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Number of columns in A = 2

Rank of A = 2

Thus, nullity = 0
- This implies that the null space of the matrix A does not contain any vectors
- Thus we can claim that all the attributes are linearly independent

R Code

```
A=matrix(c(1,3,5,2,4,6),ncol=2, byrow=F)
columns=ncol(A)
library(pracma)
rank=Rank(A)
nullity=columns-rank
```

Console output

```
> A
     [,1] [,2]
[1,]  1   2
[2,]  3   4
[3,]  5   6

Console output  


```
> print(columns)
[1] 2
> print(rank)
[1] 2
> print(nullity)
[1] 0
```


```

Linear Algebra
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Let us take some examples to make these ideas little more concrete. Let us take a matrix A which is 1, 3, 5, 2, 4, 6. A quick look at this matrix and the numbers would tell you that these two columns are linearly independent and subsequently because these columns are linearly independent there can be no relationships among these two variables.

So, you can see that the number of columns 2 since they are independent the rank is 2. Since the rank is 2, nullity is 0 and because both the variables are independent you cannot find a relationship. If you were able to find a relationship then the rank should not have been 2. So, this basically implies that null space of the matrix A does not contain any vectors and as we mentioned before these variables are linearly independent.

Now, if you want to do the same thing in R what you do is you define the matrix A which basically is done using this command. This n

columns equal to 2 tells you how many columns this number should be put in. So, since there are two columns, these numbers will be partitioned into 1, 3, 5, 2, 4, 6 and as we saw before you can actually get the rank of A.

And you can print the number of columns, you can print the rank and you can print nullity which is the difference between columns and the rank number of columns and the rank.

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Null space: Another example

- Now consider A with attributes $\{x_1, x_2, x_3\}$ such that

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 3 & 6 & 1 \end{bmatrix}$$

Number of columns in A = 3

Rank of A = 2

Thus, nullity = 1
- Thus, we need to identify the vectors in the null space of A which is non-zero in this case

R Code

```
A=matrix(c(1,2,3,2,4,6,0,0,1),ncol=3, byrow=F)
columns=ncol(A)
library(pracma)
rank=Rank(A)
nullity=columns-rank
```

Console output

```
> columns
[1] 3
> rank
[1] 2
> nullity
[1] 1
```



Linear Algebra

Now, let us take the other example that we talked about where I mentioned that we have deliberately made the second column twice the first column. So, in this case as we saw before the rank of the matrix would be 2 because there are only two linearly independent columns and since the number of variables = 3, nullity will be $3 - 2 = 1$.

So, when we look at the null space vector you will have one vector which will identify the relationship between these three variables.

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Null space: Further Example

$$A\beta = 0$$

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 4 & 0 \\ 3 & 6 & 1 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

• Thus we obtain,

$$\begin{aligned} b_1 + 2b_2 &= 0 \\ b_3 &= 0 \end{aligned}$$

• The null vector is $B = [b_1 \ b_2 \ b_3]^T = [-2b_2 \ b_2 \ 0]^T = k[-2 \ 1 \ 0]^T$

• We see that we obtain a direct linear relationship between the attributes of A using null space and rank-nullity theorem

• The same concept can be extended for bigger data set

Handwritten notes: $A\beta = 0$, CP , $A\beta = 0$, $C\beta = 0$, $b_1 = -2b_2$, $-2x_1 + x_2 = 0$

Linear Algebra 27

So, to understand how to calculate the null space let us look at this example. So, we set up this equation $A\beta = 0$ and we know we will get only 1 β here and β we have written as $b_1 \ b_2 \ b_3$ and when we do the first row versus column multiplication I will get $b_1 + 2b_2 = 0$.

When I do the second row and column multiplication I will get $2b_1 + 4b_2 = 0$ and when you do the third multiplication you get $b_3 = 0$.

Now the second equation which is $2b_1 + 4b_2 = 0$ is simply twice the first equation. So, that does not give me any extra information so, I have dropped that equation. Now, when you want to solve this notice that b_3 is fixed to be 0.

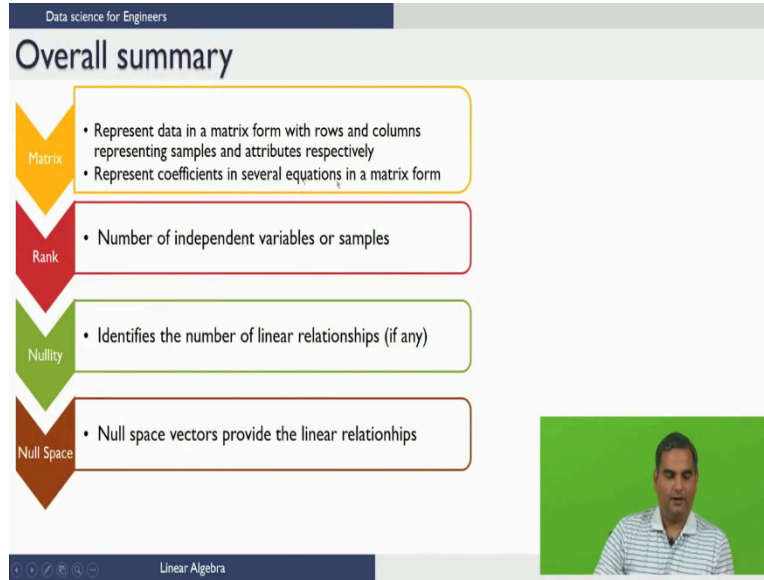
However, from this equation what you can get is b_1 is $-2b_2$. So, what we have done is instead of b_1 we have put $-2b_2$ retain b_2 and

This basically tells us that you can get a null space vector which is $-2 \ 1 \ 0$; however, whatever scalar multiple you use, it will still remain a null space vector.

So, this is easily seen from the following if A times $\beta = 0$ where β is a vector A is a matrix. Let us assume I take some other vector from β which is some $C\beta$, where C is a constant. Then if I plug it back in I will get A times $C\beta = 0$ which because this is scalar I can take it out $C A\beta = 0$. Since, this is $0 \ C$ times 0 will be 0 . So, this will be a also a 0 vector.

So, whenever β is a null space vector then any scalar multiple of that will also be a null space vector that is what is seen by this k here. Nonetheless we have a relationship between these variables which is basically saying $-2x_1 + x_2 = 0$ is a relationship that we can get out of this null space vector.

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Overall summary

- Matrix**
 - Represent data in a matrix form with rows and columns representing samples and attributes respectively
 - Represent coefficients in several equations in a matrix form
- Rank**
 - Number of independent variables or samples
- Nullity**
 - Identifies the number of linear relationships (if any)
- Null Space**
 - Null space vectors provide the linear relationships

Linear Algebra

So, to summarize this lecture as we saw matrix can be used to represent data in rows and columns; representing samples and variables respectively. Matrices can also be used to store coefficients in several equations which can be processed later for further use. The notion of rank, gives you the notion of number of in-dependent variables or samples. The notion of nullity identifies the number of linear relationships, if any between these variables and the null space vectors actually give us the linear relationships between these variables. I hope this lecture was understandable and we will see you again in the next lecture.

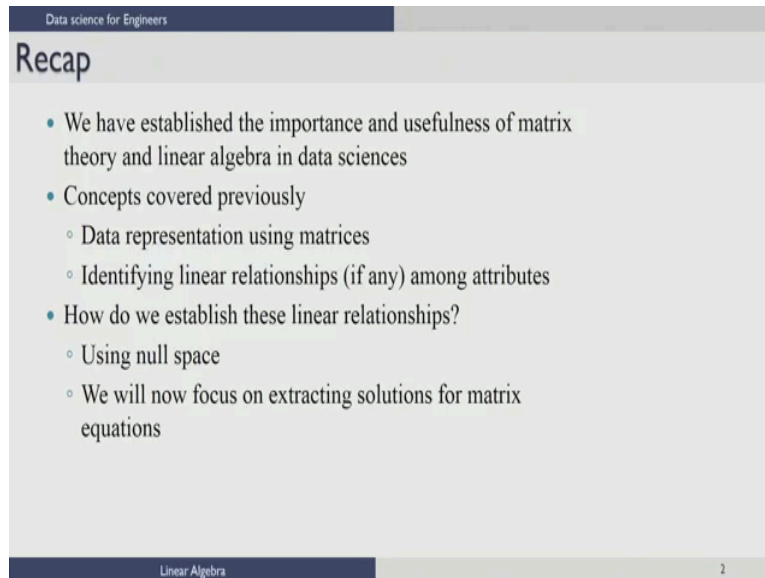
Thank you.

Data Science for Engineers
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Lecture – 13
Solving Linear Equations

In this lecture, we will discuss solutions to linear equation. As we saw in the previous lecture we had established the importance of matrix theory and linear algebra in data science.

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The slide is titled "Recap" and is part of a presentation on "Data science for Engineers". It lists the following points:

- We have established the importance and usefulness of matrix theory and linear algebra in data sciences
- Concepts covered previously
 - Data representation using matrices
 - Identifying linear relationships (if any) among attributes
- How do we establish these linear relationships?
 - Using null space
 - We will now focus on extracting solutions for matrix equations

The slide footer indicates "Linear Algebra" and the number "2".

The concepts that we covered previously are data presentation using matrices and from matrix of data, we saw how to identify linear relationships if any, among the variables or attributes. We saw that we could establish these linear relationships using the concept of null space.

In this lecture we will focus on the next important topic of extracting solutions for matrix equations, which is the most fundamental aspect of data science, where we might have several equations and we might have to find solutions to those equations. So, we will look at solving matrix equations.

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Data science for Engineers
Linear Algebra

Preliminaries

- We consider the following set of equations

$$Ax = b$$

$$A(m \times n); x(n \times 1); b(m \times 1)$$
- Generalized linear equations can be represented in the above format.
- m and n are the number of equations and variables respectively.
- b is the general RHS commonly used



In general when we have a set of linear equations, when we write it in the matrix form, we write this in the form $Ax = b$ where A is generally a matrix of size m by n , and as we saw in the last class m would represent the number of rows and n would represent the number of columns, and for matrix multiplication to work, x has to be of size n by 1 and b has to be of size m by 1 .

Now if you take each row of this equation you will have a left hand side and the right hand side. And the left-hand side will have terms corresponding to multiplying the first row of A with x and the right-hand side will have the term corresponding to b . If you take the first row, it will be the first equation and so on. So, from that viewpoint, m represents the number of equations in the system of equations and n represents the number of variables, and in general b is the constant matrix that is used on the right hand side. So, when we write $Ax = b$, this represents a set of m equations in n variables.

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Categorization

$m = n$	• Number of equations and variables are the same • Easiest case to solve
$m > n$	• More equations than variables • Usually no solution
$m < n$	• Number of equations less than number of variables • Usually multiple solutions

We look into these cases independently

Linear Algebra



Now, clearly there are three cases that one needs to address when $m = n$; that means, the number of equations and variables are the same. So, this turns out to be the easiest case to solve. When m is greater than n ; that means, we have more equations than variables. So, we might not have enough variables to satisfy all the equations. In the usual case this will lead to no solution when m is less than n .

The number of equations are less than the number of variables. What this basically means, is that we have lot more variables than necessary to solve the given set of equations. So, in a general case this will usually lead to multiple solutions. So, the first case is the easiest to solve the second case does not have solution usually, and the third case has multiple solutions. What we are going to do, is we are going to look into these cases independently, and then combine all of them using the concept of pseudo inverse.

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Full row and column rank: Concepts

- Consider a matrix data matrix $A (m \times n)$

Full Row Rank	Full Column Rank
- When all the rows of the matrix are linearly independent - Data sampling does not present a linear relationship – samples are independent	- When all the columns of the matrix are linearly independent - Attributes are linearly independent

Row rank = Column rank

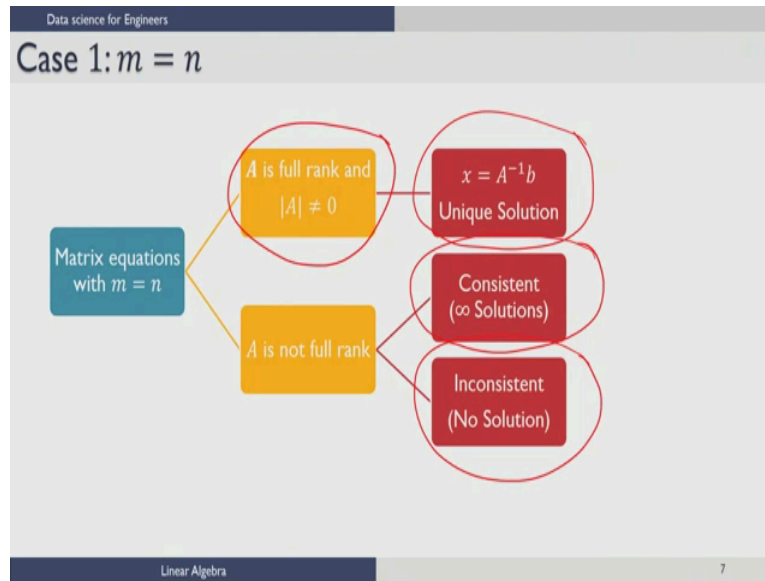
Linear Algebra



We had already discussed the concept of rank in the previous lecture, but let us talk about it again, because this is going to be useful, as we talk about solving equations. If you can consider a matrix A m by n . Now if all the rows of the matrix are linearly independent, then remember we said the rows represent data. So, this, this basically means that the data sampling does not present a linear relationship; that is the samples are independent of each other.

Now when all the columns of the matrix are linearly independent that basically means the variables are linearly independent, because columns represent variable. In a general case if I have a matrix m by n , if m is smaller than n , the maximum rank of the matrix can only be m . So, the maximum rank can be the less of the two numbers. So, in cases where I have A matrix m by n , where n is smaller than m , then the maximum rank that is possible is n . In general whatever be the size of the matrix, it has been established that row rank = column rank. So, you cannot have a different row rank and column rank. What it basically means is, whatever be the size of the matrix, if you have a certain number of independent rows, you will have only those many numbers of independent columns and so on. So, this is something important to bear in mind.

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Now, let us look at the case of $m = n$; that is we have the same number of equations and variables. If A is full rank, what does full rank mean? Now we have the same number of equations and variables $m = n$. So, the rank of the maximum rank of the matrix can be m or n , because both are the same. Now if the rank of the matrix turns out to be m , then it is what is called the full rank matrix, what this basically means, that all of these equations on the left hand side are independent of each other. In other words you can never get any equation on the left hand side as a linear combinations of other equations on the left hand side. In this case there is a unique solution to the $Ax = b$ problem, and that unique solution is $x = A^{-1}b$.

Now from your high school and so on, you would have learned that if the determinant is not 0, A^{-1} is possible to compute. So, one could simply compute $x = A^{-1}b$ as a solution to this problem, the difficulty arises only when A is not fulltime; that means, the rank of the matrix is less than n . In this case what it means, is if I take the left-hand side of the equation $Ax = b$, and then make some linear combinations of some rows of A , at least one of the rows of A is going to be a linear combination of the other rows of A ; that is the reason why the rank of the matrix became less than m . In this case, depending on what the values are on the right hand side, you could have two situations; one situation is what we are going to call as a consistent situation.

I will explain this through an example in later slides when you have a consistent situation, then you will have infinite number of solutions. There could be many solutions for $Ax = b$. And in the case where the system of equations become inconsistent, there will be no solution to this problem. So, to summarize when m equal n this is what is called a

square matrix, and if the matrix is full rank determinant $A \neq 0$, then there is a unique solution $x = A^{-1} b$, and when A is not full rank there are two situations that are possible! One is what we call as a consistent scenario, where we could have infinite solutions, and the other one is what is called the inconsistent scenario where we might have no solution.

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Case 1: Example 1.1

$$Ax = b \quad \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 7 \\ 10 \end{bmatrix}$$

$|A| \neq 0$
 $\text{rank}(A) = 2 = \text{no. of columns}$

- This implies that A is full rank

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} -2 & 1.5 \\ 1 & -0.5 \end{bmatrix} \begin{bmatrix} 7 \\ 10 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (1, 2)$

R Code

```
A=matrix(c(1,2,3,4),ncol=2, byrow=F)
b=c(7,10)
x=solve(A)%*%b
```

Console output

```
> x
     [,1]
[1,] 1
[2,] 2
```

Linear Algebra



Let us take a simple example where I have on the top of the screen, the matrix in the form $Ax = b$. In this case matrix A is $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$ x is $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ b is $\begin{bmatrix} 7 \\ 10 \end{bmatrix}$. So, you can notice that there are two equations in two variables x_1 and x_2 . The first equation is basically $x_1 + 3x_2 = 7$ and the second equation is $2x_1 + 4x_2 = 10$. We can see that this is full rank, because whatever multiple of the first column you take, you can never represent the second column and similarly whatever multiple of the first row you take you can never represent the second row, and this can also be seen from the fact that the determinant of A is not 0.

From your high school, you know the determinant is going to be in this case simply $4 \times 1 - 2 \times 3$. So, this is not 0. So, rank of A is 2. So, we have maximum rank that matrix size is 2 by 2, and the rank is 2. So, this implies that the matrix is full rank. Now we said in the previous slide that $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ can be written as $A^{-1}b$. So, this is A^{-1} matrix $\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}^{-1}$, this inverse b . You can do an inverse computation and then calculate the solution as $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and if we put the solution back into the equation, you will see $1 \times 2 + 3 \times 1 = 7$ and $2 \times 1 + 4 \times 2 = 10$. So, the solution $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ satisfies this equation.

Now if you want to write an r code for this. It is very simple, you write a matrix put the numbers in c and then define the number of

columns, you define what b is and then simply use the command solve for x to get the solution. And as you notice here the solution is 1 2. So, this is a case of full rank where I get an unique solution. The important thing to note here is, no other solution will be able to satisfy these two equations.

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Data science for Engineers

Case 1: Example 1.2

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$|A| = 0$; $\text{rank}(A) = 1$; nullity = 1

- Checking consistency

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$$

$$\text{Row}(2) = 2\text{Row}(1)$$
- The equations are consistent with only one linearly independent equation
- The solution set for (x_1, x_2) is infinite because we have only one linearly independent equation and 2 variables

Handwritten notes:

$$x_1 + 2x_2 = 5$$

$$x_2 = 0, x_1 = 5$$

$$x_2 = 1, x_1 = 3$$

Linear Algebra

Now, let us take another example to illustrate what happens when the rows or columns become linearly dependent. So, here is another set of equations and we have to read these equations as $x_1 + 2x_2 = 5$ and $2x_1 + 4x_2 = 10$.

Now if you notice these equations, through the matrix A you will see that, if I multiply the first column by 2 I get the second column. So, the second column is linearly dependent on the first column or the first column is linearly dependent on the second column, whichever way you want to say it. Similarly if you divide for example, the second row by 2 you will get the first row or if you multiply the first row by 2 you get the second row. So, the rows are also linearly dependent and, and as I said before, there is only one independent column and that necessarily means that there will be only one linearly independent row.

Now another way to check this linear dependence is to calculate the determinant of this matrix, and when we calculate the determinant of this matrix, you will get 1 times 4 - 2 times 2, which is 4 - 4 which is 0. So, determinant also says there is linear dependence here. And from the previous lecture we know that when the rank is 1 and the number of columns are 2 the nullity is 1. So, there is one vector in the null space. Now let us look at the equations when you write these equations, as I

said before the first equation $x_1 + 2x_2 = 5$, and the second equation is $2x_1 + 4 = 4x_2 = 10$.

When we talk about the linear dependence of the rows of A , we are only talking about the left hand side, we never talked about the right hand side. Now whenever the left hand side becomes linearly dependent, if the same linear dependence is maintained on the right hand side also then we have the situation of consistent equations. So, if you take a look at this example, we know the left hand side, if I take the first equation and multiply it by 2 I get the second equation on the left hand side. So, $x_1 + 2x_2$ multiplied by 2 gives me $2x_1 + 4x_2$. Now if the same linear dependence is maintained on the right hand side; that is if I take the first number 5 and multiply it by 2 I should get this number.

In this case, we have constructed this example in such a way that we get this number. Now not only is the left hand side linearly dependent, the same linear dependence is also maintained on the right hand side. So, as a whole the equations become consistent, but linearly dependent on each other. So, in this case you notice that if I solve this equation. I do not have to solve this equation, because I multiply this by 2 I get this equation. So, whatever 1 and x_2 will solve this equation will also solve the second equation. So, I can drop one of these two equations. Let us assume I drop this equation out. So, I am just left with $x_1 + 2x_2 = 5$, but now notice that I have one equation in two variables, that basically tells me I have one free variable. So, for example, if I take this equation $x_1 + 2x_2 = 5$.

If I set $x_2 = 0$ I get $x_1 = 5$. If I said $x_2 = 1$ I get $x_1 = 3$ and so on. So, all of these are solutions to these equations. I am just pointing out two. Now you can notice that I can take any value for x_2 and then calculate an x_1 , which will satisfy this equation. Since I can take any value for x_2 . There are infinite choices for x_2 corresponding to each one of these infinite choices, I will get a value of x_1 . So, that pair would be a solution to this set of equations. So, when this A becomes rank less than full rank, if the equations are consistent then they will get infinitely number of many infinitely many solutions.

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Data science for Engineers

Case 1: Example 1.3

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 9 \end{pmatrix}$$

$$|A| = 0$$

$$\text{rank}(A) = 1$$

$$\text{nullity} = 1$$

- Checking consistency

$$\begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 4x_2 \end{bmatrix} = \begin{pmatrix} 5 \\ 9 \end{pmatrix}$$

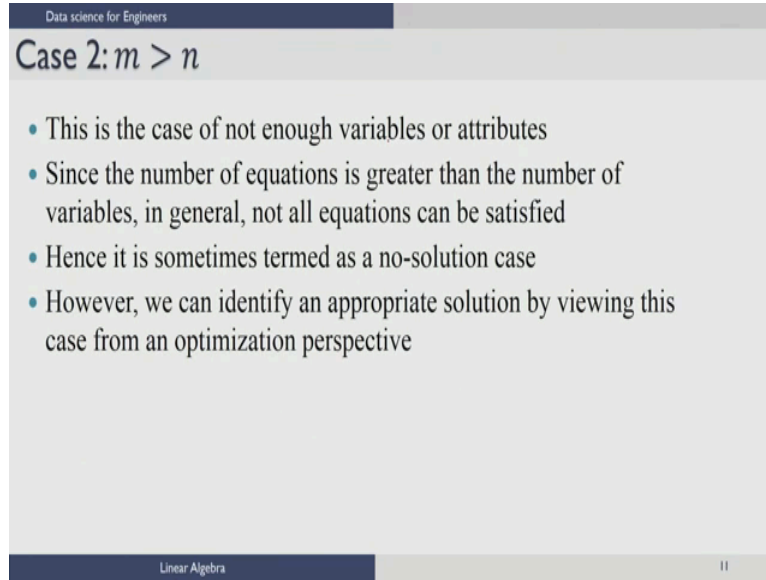
$$2\text{Row}(1) = 2x_1 + 4x_2 = 10 \neq 9$$
- Thus the equations are inconsistent
- One cannot find a solution to (x_1, x_2)

Linear Algebra 10

Let us take another example to explain the idea of inconsistent equations. Let us look at this example here, this is the same example as the previous case except for one change. This number has been changed from 10 to 9. In the previous case we saw that the left hand side, if I multiply the first equation by 2, I get the second equation. And since in the previous example this number was 10, when I multiplied 5 by 2 I get 10. So, both the equations became consistent and I could drop one equation ; however, if this number was anything other than 10 then what it basically means is, that while the left hand side will be linearly dependent where the second equation will be twice the first equation the right hand side will always be inconsistent.

If this number is not there that basically means whatever x_1 and x_2 values that you use for solving the first equation, If you plug that value into the second equation, your right hand side will always be 10, but you want it to be 9; so which is never possible. So, this is a case where you will not have any solution. So, in other words if this number is anything other than 10, you will have no solution to these equations, this will become a inconsistent case. Now again inconsistent case is possible, only when the rank of the matrix is less than full rank. That is there are some rows which are linearly dependent on the left hand side, which you can again verify by the same determinant $1 \times 4 - 2 \times 2$ is 0. And since the second row is inconsistent, these equations are inconsistent and one cannot find a solution to this problem of $Ax = b$.

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Data science for Engineers

Case 2: $m > n$

- This is the case of not enough variables or attributes
- Since the number of equations is greater than the number of variables, in general, not all equations can be satisfied
- Hence it is sometimes termed as a no-solution case
- However, we can identify an appropriate solution by viewing this case from an optimization perspective

Linear Algebra 11

So, that finishes the case where the number of equations and variables are the same. So, till now we saw the case, where $m = n$. Now let us take the second case, where m is greater than n . Since m is greater than n this basically means that I have more equations than variables. So, this is the case of not enough variables or attributes to solve all the equations. Since the number of equations is greater than the number of variables in general, we will not be able to satisfy all the equations; hence we termed this as a no solution case.

However we still want to identify some solution which makes sense and which we can generalize for all the three cases. So, what we are going to do is, we are going to identify an appropriate solution by viewing this case from an optimization perspective and I explain what that is presently.

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Data science for Engineers

Case 2: An optimization perspective

- Instead of identifying a solution to $Ax - b = 0$, one can identify an x such that $(Ax - b)$ is minimized
- Notice that $(Ax - b)$ is a vector
- There will be as many error terms as the number of equations
- Denote $(Ax - b) = e(mx1)$; there are m errors $e_i, i = 1:m$
- One could minimize all the errors collectively by minimizing $\sum_{i=1}^m e_i^2$
- This is the same as minimizing $(Ax - b)^T(Ax - b)$

Handwritten notes on the slide:

$$\begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix}^T \begin{pmatrix} e_1 \\ e_2 \\ e_3 \end{pmatrix} = e_1^2 + e_2^2 + e_3^2$$

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{21}x_1 + a_{22}x_2 &= b_2 \\ a_{31}x_1 + a_{32}x_2 &= b_3 \end{aligned}$$

Linear Algebra



Let us look at a solution to $Ax = b$ when the number of equations are more than the number of variables. As we mentioned before we are going to take an optimization perspective here. When we try to solve $Ax = b$, we can write that equation as $Ax - b$, and if there is a perfect solution to the set of equations then $Ax - b$ will be $= 0$. However, since we know that the number of equations are a lot more than the number of variables, there might not be a perfect solution.

So, what we want to do is, we want to identify a solution in such a way that $Ax - b$ is minimized. Why do we want to do this? Notice that $Ax - b$ is a vector and if you take each term in that vector you can think of each of those terms as an error in an equation. So, to give you an example, if I have $a_{11}x_1 + a_{12}x_2 = b_1$, $a_{21}x_1 + a_{22}x_2 = b_2$, $a_{31}x_1 + a_{32}x_2 = b_3$. Now if I had a perfect solution x_1, x_2 which will satisfy all the three equations.

Then when I write this as $a_{11}x_1 + a_{12}x_2 - b_1$, this will be $= 0$; however, when I cannot find a perfect solution then let me call this as an error. In this equation correspondingly I allow an error in this equation, I have error in this equation. So, you notice that there are three errors - as many errors as there are equations. So, how do we collectively minimize all of these errors? One thing to immediately think of both is to minimize $e_1 + e_2 + e_3$, but that would not be a good idea simply, because I could have a_1 as a very large error in the positive direction e_2 as a very large error in the negative direction and e_3 as 0 that will still give me an answer 0. So, that is not a good answer at all, one way to do this is to collectively minimize all of them by minimizing what we call as the sum of squares errors. So, you take this example instead of minimizing $e_1 + e_2 + e_3$. You are going to minimize $e_1^2 + e_2^2 + e_3^2$. In this case notice irrespective of whether e_1, e_2, e_3 are positive or negative as long as they are away from 0. The contribution

to the error term will be high. So, it will automatically ensure that you do not go very far away from zero.

Now, this is the least squares solution you could also minimize instead of e_1^2 , you can minimize modulus $e_1 + \text{mod } e_2 + \text{mod } e_3$, because mod is always positive; that is also possible, but in general we are going to talk about least square solution where we minimize this sum of squares of errors. Now this is the same as minimizing $\|Ax - b\|$ simply, because $Ax - b$ is e . So, $(Ax - b)^T$ is e^T . So, if I have numbers $e_1 \ e_2 \ e_3^T$ and multiply by $e_1 \ e_2 \ e_3$, this will lead to $e_1^2 + e_2^2 + e_3^2$ which is the same as this right here. So, minimizing this, is the same as minimizing $(Ax - b)^T (Ax - b)$.

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Data science for Engineers

Case 2: An optimization perspective

- This optimization problem is

$$\begin{aligned}
 &\min[(Ax - b)^T (Ax - b)] \\
 &= \min[(b^T - x^T A^T) (Ax - b)] \\
 &= \min[(x^T A^T Ax - 2b^T Ax + b^T b)] = f(x)
 \end{aligned}$$
- We observe that the optimization problem is a function of x
- Solving the optimization problem will result in a solution for x
- The solution to this optimization problem is obtained by differentiating $f(x)$ with respect to x and setting the differential to zero

$$\nabla f(x) = 0$$



Linear Algebra

So, the optimization problem is minimize $(Ax - b)^T (Ax - b)$. After some algebraic manipulation we can write this objective as a function of the solution $f(x)$. We observe that this optimization problem becomes a problem, where the objective is a function of x . Solving this optimization problem will result in a solution for x , which is what we are going for. So, the way to get the solution to deck this optimization problem, is to take $f(x)$ differentiate it with respect to x and set it to 0.

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Data science for Engineers

Case 2: An optimization perspective

- Differentiating $f(x)$ and setting the differential to zero results in

$$2(A^T A)x - 2A^T b = 0$$

$$(A^T A)x = A^T b$$
- Assuming that all the columns are linearly independent

$$x = (A^T A)^{-1} A^T b$$



Linear Algebra

So, differentiating $f(x)$ and setting the differential to 0 results in the following equation, and this can be simplified to this equation, where A^T a times x is $A^T b$. Now to solve this equation we will assume that all the columns are linearly independent which allows us to take the inverse of this matrix, and then we can come up with a solution x equal $(A^T A)^{-1} A^T b$. While this solution x might not satisfy all the equations, this solution will ensure that the errors in the equations are collectively minimized. So, this is an optimization view for case 2, where the number of equations are more than the number of variables or m is greater than n .

We will conclude this lecture at this point. What I will do in the next lecture is, take an example to illustrate what happens in case 2 in terms of how you get a solution and whether some of these equations are satisfied or not satisfied and so on, and after that I will move on to case₃ and show an optimization perspective for solving those types of equations, where the number of variables become greater than the number of equations, and then in the next lecture I will also show how all of this can be combined into one elegant solution through the concept of pseudo inverse.

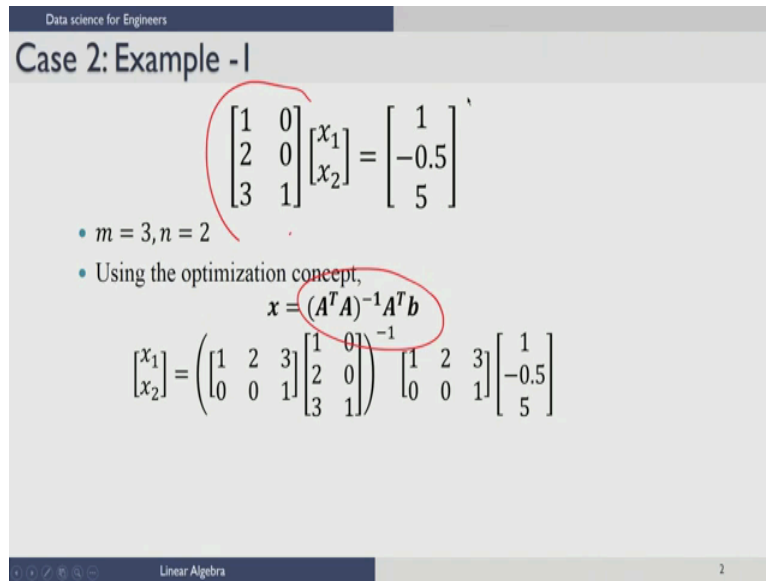
Thank you.

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Lecture – 14
Solving Linear Equations

We will continue the lecture on solving linear equations. In the last lecture I discussed the case of many more equations and variables, where we might not have a solution and how we can use an optimization perspective to find a solution. In this lecture I am going to give you some examples for that case show you what happens when we apply the solution that we derived last time, and then after that I will go on to look at the case of more variables than equations.

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Case 2: Example - I

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

- $m = 3, n = 2$
- Using the optimization concept,

$$x = (A^T A)^{-1} A^T b$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Linear Algebra

So, let us look at an $Ax = b$ example system as shown in the screen. Here we have a matrix with 3 rows and 2 columns, which basically means that there are 3 equations in 2 variables number of equations more than number of variables. And we have to read these equations as $x_1 = 1$, $2x_1 = -0.5$ and $3x_1 + x_2 = 5$.

So, if you notice these equations you would realize that the first 2 equations are inconsistent. For example, if we were to take the first equation is true then $x_1 = 1$ and if we substitute that value into the

second equation you will get $2 = -0.05$. If you were to take the second equation as true then $2x_1$ is -0.5 . So, x_1 will be -0.25 and that would not solve the first equation. So, these 2 equations are inconsistent. The third equation since it is $3x_1 + x_2$ irrespective of whatever value you get for x_1 you can always use this equation to calculate the value for x_2 ; however, we cannot solve this set of equations.

Now, let us see what is the solution that we get, by using the optimization concept that we described in the last lecture. We said $x = A^T A \text{ inverse } A^T b$. the A matrix is $1 \ 0 \ 2 \ 0 \ 3 \ 1$. So, A^T matrix is $1 \ 2 \ 3 \ 0 \ 0 \ 1$. Simply plugging in the matrices here.

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Case 2: Example continued

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 15 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 5 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (0, 5)$
- Substituting in the equation shows

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 5 \end{bmatrix} \neq \begin{bmatrix} 1 \\ -0.5 \\ 5 \end{bmatrix}$$

Linear Algebra 3

And then doing the calculation gives us this equation which says x_1 x_2 is a matrix times $15 \ 5$. This is an intermediate step for the calculation. And when you further simplify it you get a solution $x_1 = 0$, $x_2 = 5$. Notice that the optimum solution here that is chosen does not have either one of the 2 cases that we talked about in the last slide, which is $x_1 = 1$ and $x_1 = -0.25$ the optimization approach chooses $x_1 = 0$ and $x_2 = 5$ and when you substitute it back into the equation you get b as $0 \ 0 \ 5$ whereas, the actual b that we are interested in is $1 \ -0.55$.

So, you can see that while the third equation is being solved exactly the first two equations are not solved for; however, as we described before this is the best solution in a collective minimization of error sense, which is what we defined as minimizing sum of squared of errors. We will now move on to the next example.

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Case 2: Example

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

- $m = 3, n = 2$
- Using the optimization concept,

$$x = (A^T A)^{-1} A^T b$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$


Linear Algebra

Let us consider another example for us to illustrate something different here. We have taken the same left hand side we have the same a matrix; however, the right-hand side has been modified to be 1 2 5. we have done this for a specific reason which we will see presently. So, when you look at this equation; if you take the first equation it reads as $x_1 = 1$. If you look at the second equation it reads as $2x_1 = 2$. The third equation reads as $3x_1 + x_2 = 5$.

So, from the first equation you can get a solution for $x_1 = 1$ and the second equation since it reads as $2x_1 = 2$, we have to simply substitute the solution that we get from the first equation and see whether the second equation is also satisfied since $x_1 = 1$ 2 times x_1 2 times 1 is 2 the second equation is also satisfied.

Now, let us see what happens to the third equation. The third equation reads as $3x_1 + x_2 = 5$, we already know $x_1 = 1$ satisfies the first 2 equations. So, $3x_1 + x_2 = 5$ would give you $x_2 = 2$. Now you notice that if I get a solution 1 and 2 for x_1 and x_2 ; though the number of equations are more than the variables, the equations are in such a way that I can get a solution for x_1 and x_2 that satisfies all the 3 equations.

Now let us see whether the expression that we had for this case actually uncovers this solution. So, we said $x = (A^T A)^{-1} A^T b$ and we do the same manipulation as the last example except that this b has become 1 2 5 now.

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Case 2: Example continued

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0.2 & -0.6 \\ -0.6 & 2.8 \end{bmatrix} \begin{bmatrix} 20 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Thus, the solution for the given example is $(x_1, x_2) = (1, 2)$
- Substituting in the equation shows

$$\begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

R Code

```
A=matrix(c(1,2,3,0,0,1),ncol=2, byrow=F)
b=matrix(c(1,2,5),ncol=2, byrow=F)
x=inv(t(A)%*%A)%*%t(A)%*%b
x
```

Console output

```
> x=inv(t(A)%*%A)%*%t(A)%*%b
> x
[1]
[1,] 1
[2,] 2
>
```



Linear Algebra

After some more calculations you will see that $x_1 = 1$ $x_2 = 2$. Thus, the solution is 1 2 and we had already verified that this would solve the equation and we had verified that 1 2 is a solution that we can directly get by observation from the previous slide.

So, the important point here is that if we have more equations than variables then you can always use this least square solution which is $(A^T A)^{-1} A^T b$. The only thing to keep in mind is that $(A^T A)^{-1}$ exists if the columns of A are linearly independent. If the columns of A are not linearly independent, then we have to do something else which you will see as we go through this lecture.

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Case 3: $m < n$

- This case addresses the problem of more attributes or variables than equations
- Since the number of attributes is greater than the number of equations, one can obtain multiple solutions for the attributes
- This is termed as an infinite-solution case
- How does one choose a single solution from the set of infinite possible solutions?



Linear Algebra

So, that finishes the case where the number of equations are more than the number of variables. Now let us address the last case where the number of equations are less than the number of variables, which would be m less than n in this case we address the problem of more attributes or variables than equations.

Now since I have many more variables and equations I would have infinite number of solutions the way to think about this is the following. If I had, let us say, 2 equations and 3 variables. You can think of this situation as one where you could choose any value for x_3 and then simply put it into the 2 equations. And whatever are the terms with respect to x_3 you collect them and take them to the right-hand side; that would leave you with 2 equations and 2 variables and once we solve for that 2 equations and 2 variables we will get values for x_1 and x_2 .

So, basically what this means is that, I can choose any value for x_3 and then corresponding to that I will get values for x_1 and x_2 . So, I will get infinite number of solutions. Since I have infinite number of solutions then the question that I ask is how do I find one single solution from the set of infinite possible solutions? Clearly if you are looking at only solvability of the equation, there is no way to distinguish between this infinite possible solutions. So, we need to bring some other metric that we could possibly use, which would have some value for us to pick one solution that we can say is a solution to this case.

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Linear Algebra

Case 3: An optimization perspective

- Pose the following optimization problem

$$\min \left(\frac{1}{2} x^T x \right) \text{ s.t. } Ax = b$$
- Define a Lagrangian function $f(x, \lambda)$

$$\min \left[f(x, \lambda) = \frac{1}{2} x^T x + \lambda^T (Ax - b) \right]$$
- Differentiating the Lagrangian with respect to x , and setting to zero

$$x + A^T \lambda = 0$$



Similar to the previous example we are going to take an optimization view here, what we are going to do is we are going to minimize $x^T x$, this half is just to make sure the solution comes out in a nice form. And notice here something that is important we also have a constraint for this optimization problem $s \cdot t$ means subject to.

So, I want to minimize this half; $x^T x$ subject to the constraint $Ax = b$. So, in other words what we are saying is whatever solution we get for x that has to necessarily satisfy this equation. And this is not a problem we can find infinite number of solutions x which will satisfy these equations. So, what this objective does is of all of those solutions how do I pick, that one solution which will minimize this $x^T x$. We have to think about a rationale for, why we would choose $x^T x$ as an objective.

This basically says that of all the solutions I want the solution which is closest to the origin is what this is saying in terms of $x^T x$. From an engineering viewpoint one could justify this as the following; if you have lots of design parameters that you are trying to optimize and so on, you would like to keep the sizes small for example, so you might want small numbers. So, you want to be as close to origin as possible this is just one justification for doing something like this nonetheless this is one way of picking one solution from this infinite number of solutions.

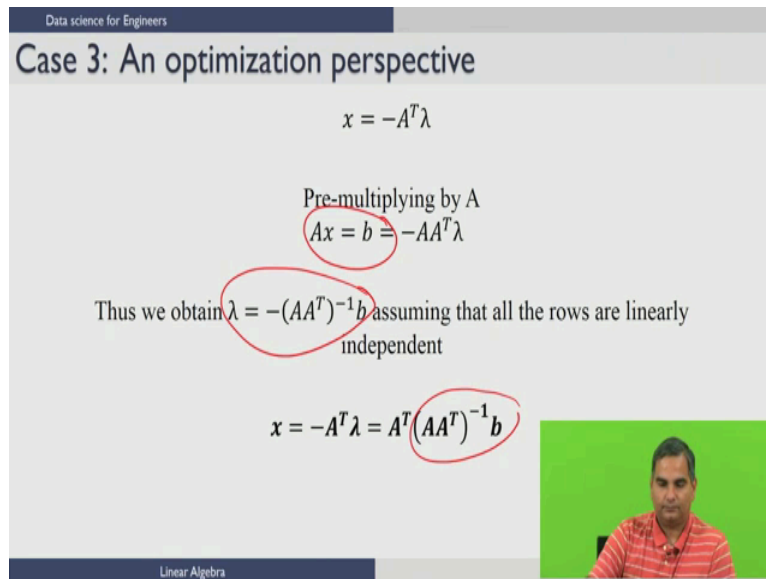
Now, in the previous example and in this example, we are solving these optimization problems; however, we have not taught in this course how to solve optimization problems. For people who already know how to solve optimization problems this would be obvious. For other participants who do not know how to solve optimization problems, I would encourage you to just bear with me and then go through this solution and see what the solution form is and once this module on linear algebra is finished we will have a couple of modules on optimization from the viewpoint of data science.

So, when we do that, you will see how we solve these kinds of optimization problems. The optimization problem that we solved for the last case is what is called an unconstrained optimization problem because there are no constraints to that problem whereas, this problem that we are solving is called a constrained optimization problem because while we have an objective we also have a set of constraints that we need to solve.

So, you will have to bear with us till you go through the optimization module to understand this. Interestingly it is generally a good idea to teach linear algebra on optimization, but interestingly, some of the linear algebra concepts you can view as optimization problems and solving optimization problems requires lots of linear

algebra concepts. So, in that sense they are both coupled. In any case to solve optimization problems of this form we can define what is called a Lagrangian function $f(x) + \lambda^T (Ax - b)$, λ are extra parameters that we introduce into this optimization formulation. And what you do is you minimize this Lagrangian with respect to x to get a set of equations. And you also minimize this with respect to λ which will back out the constraint. So, whatever solution you have, has to solve both the differentiation with respect to x which should give you $x + A^T \lambda = 0$ and also differentiation with respect to λ which will simply give you $Ax - b = 0$. That would basically say that whatever solution you get, that has to satisfy the equation $Ax = b$ we will see how this is useful in identifying a solution.

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Case 3: An optimization perspective

$$x = -A^T \lambda$$

Pre-multiplying by A

$$Ax = b = -AA^T \lambda$$

Thus we obtain $\lambda = -(AA^T)^{-1} b$ assuming that all the rows are linearly independent

$$x = -A^T \lambda = A^T (AA^T)^{-1} b$$

Linear Algebra

So, let us look at this equation $x + A^T \lambda = 0$. So, from this we can get a solution for x which is $-A^T \lambda$. Now what you could do is; you do not know x and you do not know λ also. So, there has to be some way of finding out both of them. So, what we are going to do is we are going to use the knowledge that any solution that we get has to satisfy the equation $Ax = b$.

So, what we are going to do is we are going to pre-multiply this x by A . So, we pre-multiply on both sides so, we get $Ax = -AA^T \lambda$ by pre-multiplying this equation by A . Now since any solution x satisfies $Ax = b$, I can replace this Ax by b and I get this equation $b = -AA^T \lambda$ and from this equation we can get λ to be $-(AA^T)^{-1} b$. And this is possible and this inverse exists only if all the rows are linearly independent.

Now, since we have an expression for λ we can substitute that expression here and we will get $x = -A^T \lambda$ and your λ is this expression which is from here. So, this solves for x in the equation $Ax = b$. And since we use this idea here the x that we get is such that $Ax = b$ that is satisfies the original equation.

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Case 3: Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- $m = 2, n = 3$
- Using the optimization concept,

$$x = A^T (AA^T)^{-1} b$$

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

Linear Algebra



Now, let us take an example to understand this. I have an $Ax = b$ here, I have A as $1 \ 2 \ 3 \ 0 \ 0 \ 1$ and b as $2 \ 1$. So, again notice here since there are 2 equations, I have 2 rows and 3 equation 3 variables, I have 3 columns and these equations are read as $x_1 + 2x_2 + 3x_3 = 2$ and $x_3 = 1$. Now clearly when you look at this equation you will notice that $x_3 = 1$ has to be a solution. So, the question is how do I choose x_1 and x_2 , nonetheless we will use the optimization solution to actually see what happens here.

So, the optimization solution from the previous slide is the following $x = A^T (A A^T)^{-1} b$. Now A^T is $1 \ 2 \ 3 \ 0 \ 0 \ 1$ here. And this is my A and A^T again I take an inverse of this and b now is $2 \ 1$.

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Case 3: Example

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 14 & 3 \\ 3 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} -0.2 \\ 1.6 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -0.2 \\ -0.4 \\ 1 \end{bmatrix}$$

- The solution for the given example is $(x_1, x_2, x_3) = (-0.2, -0.4, 1)$

Linear Algebra

R Code

```
A=matrix(c(1,0,2,0,3,1),ncol=3)
b=c(2,1)
library(MASS)
x=t(A)%*%inv(A%*%t(A))%*%b
x
```

Console output

```
A=matrix(c(1,0,2,0,3,1),ncol=3, byrow=F)
b=c(2,1)
x=t(A)%*%inv(A%*%t(A))%*%b
x
```



And when I do some more algebra I finally get a solution to x_1, x_2, x_3 which is the following; And we had already seen that $x_3 = 1$ has to be a solution because the last equation basically said $x_3 = 1$. Now x_1 and x_2 you could have found several numbers to satisfy the first equation after you choose $x_3 = 1$ of all of these this solution says this - 0.2 - 0.4 is the minimum norm solution or this vector is the closest vector from the origin; that satisfies my equation $Ax = b$. So, I can finally, say my solution x_1, x_2, x_3 is - 0.2 - 0.41.

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Case 3: Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

- The solution for the given example is $(x_1, x_2, x_3) = (-0.2, -0.4, 1)$
- Verify this is a solution that satisfies the original equation
- This also turns out to be minimum norm solution

Linear Algebra



And you can easily verify that this satisfies the original equation since x_3 is 1, the second equation is $x_3 = 1$.

So, that gets satisfied when you look at the other equation you have one times - 0.2 + 2 times - 0.4. That will be - 0.2 - 0.81 + 3 times 1 will give you 3 - 1 = 2 which is this. So, the solution that we found satisfies the original equation and this also turns out to be the minimum norm solution as we discussed.

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Generalization

- The described cases cover all the scenarios one might encounter while solving linear equations
- Is there any form in which the results obtained for cases 1, 2 and 3 can be generalized ?
- The concept we used to generalize the solutions is called as Moore-Penrose pseudo-inverse of a matrix
- The pseudo inverse is used as follows

$$Ax = b$$

The solution becomes

$$x = A^+b$$
- Singular Value Decomposition can be used to calculate the pseudo inverse or the generalized inverse (A^+)

$x = A^{-1}b$

Linear Algebra
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So, when we have a set of linear equations we basically said that there are 3 cases that one needs look at, one case is where number of equations and variables are the same $m = n$. The second case is where the number of equations are lot more than the number of variables m greater than n . And the third case was when number of equations less than number of variables m less than n . And we saw that one case is an exact solution if it is a full rank matrix.

And if it is not a full rank matrix then you could have infinite solutions or no solutions, and interestingly the next 2 cases covers these 2 aspects when I have lot more equations than variables I have a no solution case, and when I have lot more variables than equations I have infinite solution case and since we are able to solve all the 3 we should be able to use the solution to the case 2 and 3 for the case one where the rank is not full. And depending on whether it is a consistent set of equation or inconsistent set of equation you should be able to use the corresponding infinite number of solutions or no solutions result right?

So, in some sense we understand that there should be some generalization of all of these results. So, that we can write one equation which solves all of these cases square rectangular cases and so on. So,

that is a question that we are asking, is there any form in which the results obtained from cases 1, 2 and 3 can be generalized. It turns out that there is a concept that we can use to generalize all of these, this is what is called the Moore Penrose pseudo inverse of a matrix.

So, when we typically have equations of the form $Ax = b$, we write $x = A^{-1}b$ as a solution. The generalization of this is to write x as A^+b where A^+ have used this term to denote the pseudo inverse of A . And as long as we can calculate the pseudo inverse in a fashion that is irrespective of the size of A , irrespective of whether the columns and rows are dependent or independent.

If I can write one general solution like this which will reduce to the cases that we discussed in this lecture, then that is a very convenient way of representing all kinds of solutions instead of looking at whether the number of rows are more, number of columns are more, is rank full and so on. All of them if they can be subsumed in one expression like this it would be very nice and it turns out that there is an expression like that and that expression is called the pseudo inverse.

Now, the pseudo inverse of A can be calculated using a singular value decomposition as one technique. There are many other ways of computing this, but singular value decomposition is one way of computing this. And as far as this course is concerned you just need to know that we can compute this. We do not have to really worry about how singular value decomposition is done.

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Two examples revisited

Example 2

R Code

```
A=matrix(c(1,2,3,0,0,1),ncol=2, byrow=F)
b=matrix(c(1,2,5),ncol=1, byrow=F)
library(MASS)
x = ginv(A)%*%b
```

Solution

```
> x [1]
[1,] 1
[2,] 2
```

Example 3

R Code

```
A=matrix(c(1,0,2,0,3,1),ncol=3, byrow=F)
b=c(2,1)
library(MASS)
x = ginv(A)%*%b
```

Solution

```
> x [1]
[1,] -0.2
[2,] -0.4
[3,] 1.0
```



Linear Algebra

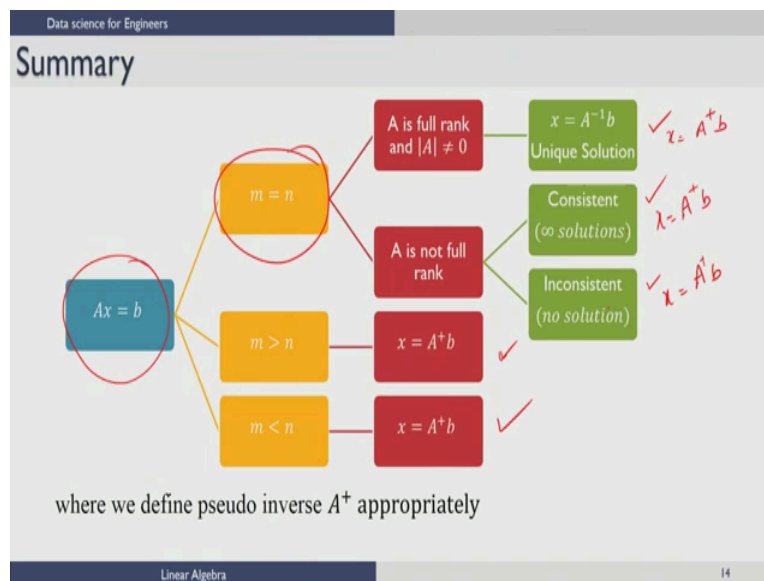
So, how do I get this in R? So the way you do this in R is you use this library and the pseudo inverse is usually calculated using this g inverse a. Here g stands for generalized. So, what R does is whatever size of the problem you give here we have given 2 different examples, where one example has more equations than variables the second example has more variables than equation.

These are the examples that were picked from this lecture itself and we show that irrespective of whatever be the sizes of this matrices a and b, we use the same equation g inverse A and the solution 1 2 that we got in one example and the solution - 0.2 - 0.4 and one we got in the other case come out of this g inverse.

Now, the key point to understand is you simply use g inverse in R to get these solutions, but the interpretation of these solutions is what we have taught in this class. So, interpretation for this solution here is that; this is the least square solution or this is the solution that will minimize the errors collectively or this is a solution that will minimize $e_1^2 + e_2^2$ and so on.

This is what is called the minimum norm solution. While there are infinite number of solutions this is a solution that is the closest to origin. So, that is the interpretation for these 2 solutions that that we want to keep in mind as far as solving linear equations is concerned, nonetheless the operationalization for how to use R is very simple you simply use g inverse as a function.

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So, let me summarize this lecture, we said we are interested in solving equations of the form $Ax = b$. We talked about 3 cases $m = n$

and m if A is full rank unique solution $A^{-1}b$. If A is not full rank there are 2 possibilities either the equations are consistent or inconsistent. And if m is greater than n we look at a least square solution and if m is less than n then we look at a least norm solution.

We can write this as $A^{-1}b$ or I could also write this as pseudo inverse b . In this case the pseudo inverse and A inverse will be exactly the same and as I mentioned before since these 2 cases are covered by these 2. I should be able to use the same a pseudo inverse b for both these cases also without worrying about whether they are consistent inconsistent and so on. In all of these cases I will get a solution by using the idea of generalized inverse.

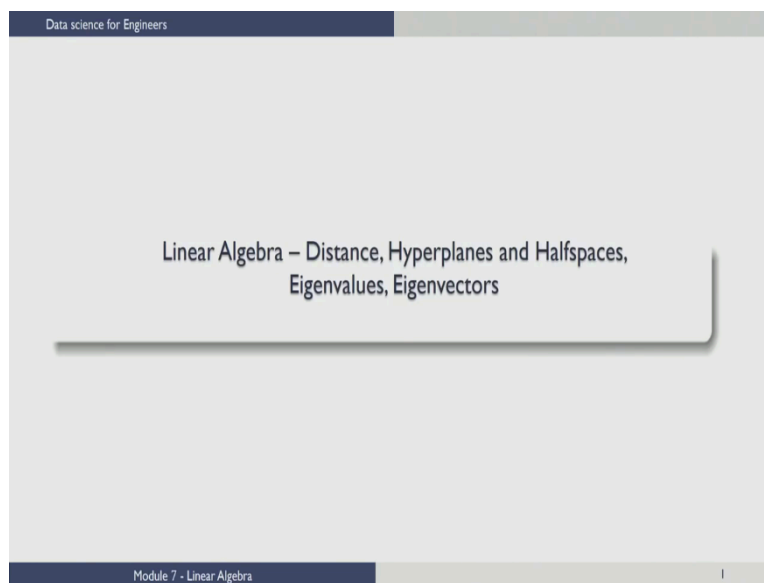
So, this concludes the section on solving linear equations irrespective of whether it is a square or a rectangular system or not, worrying about really whether the columns are dependent independent and so on. You can use generalized inverse as one unifying concept to find a solution to all these cases.

Thank you and in the next lecture we will take a geometric view of the same equations and variables that is useful in data science.

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Lecture - 15
Linear Algebra - Distance, Hyperplanes and Halfspaces, Eigenvalues, Eigenvectors

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In the previous lectures we looked at linear algebra. But we took a linear algebraic view where we looked at equations and variables and solvability of these equations and so on. The same subject we could also take a geometric view, where we think about vectors and hyperplanes, half spaces and so on. So, we are going to cover that in the next couple of lectures that we are going to have on linear algebra. While we do this, we are going to cover the ideas of distance, hyperplanes, half spaces, Eigenvalues Eigenvectors. Now, some of these are things that would be very well known to most of you nonetheless, for the sake of completeness, I will go through all of these ideas and then I will use all of those ideas, when we describe hyper planes, half spaces and so on.

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Review

- So far we have discussed linear algebra and matrix theory from a data science perspective
- We will provide some geometric interpretations now
- This section covers the following
 - Vectors
 - Notion of distance
 - Projections
 - Hyperplanes
 - Halfspaces
 - Eigenvalues and eigenvectors

Module 7 - Linear Algebra 2

So, we will cover vectors notion of distance, we will talk about projections, we will talk about hyper planes, we will talk about half spaces and then we will talk about Eigenvalues and eigenvectors in this lecture. Till now, if we have been looking at a $X = b$ and X as set of variables that needs to be calculated. So, we have been using this notation $x_1 \ x_2$ as a vector, where we have been interpreting this as a solution to a variable x_1 and a solution to variable x_2 and so on.

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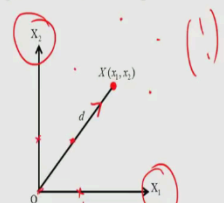
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Vectors and lengths

- Consider

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

- X is a data point in a 2 dimensional plane with x_1 and x_2 as the distances along the X_1 and X_2 axes respectively.
- X can also be considered as a vector between the origin and the data point
- The length (magnitude) of this vector is

$$d = \sqrt{x_1^2 + x_2^2}$$


Module 7 - Linear Algebra 3

Another way to think about the same vector X is to think of this as actually a point in a 2-dimensional space and here we say, it is a 2 dimensional space because there are 2 variables. So, for example, if you take x_1 and x_2 you could think of this, as being a point in a 2

dimensional space, where there is one axis that represents x_1 and there is another axis that represents x_2 , and depending on the value of the x_1 and x_2 you will have a point anywhere in this plane.

So, for example, if you have let us say 1 as your vector, and if this is one and this is one, then the point will be here and so on. So, what we are doing here is, we are looking at vectors as points in a particular dimensional space. Since, there are 2 numbers here we talked about 2-dimensional space if for example, there are 3 numbers here, then it would be a point in a 3 dimensional space, you could also think of this as a vector and we define the vector from the origin.

So, I could think of this X as a vector, where I connect origin to the point. So, this is another view of the same vector X and once we think of this as a vector then, vector has both direction and magnitude. So, in this case the direction is this and the magnitude is, what we think of as a distance from the origin and in this case we all know, this well-known formula for Euclidean distance, which is root of $(x_1^2 + x_2^2)$ right? So, that is the distance of this point from the origin.

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Vectors and lengths: Example

- Consider the point $A = (3,4)$ in a two dimensional plane

$$A = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$d = \sqrt{3^2 + 4^2} = 5 \text{ units}$$

- Important: Geometric concepts are easier to visualize in 2D or 3D
- Difficult to do so in the higher dimensions
- However, the fundamental mathematics remain the same irrespective of the dimension of the vector

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Now, just as a very, very simple example, if you have a 0.34 then you can find the distance from the origin is root of $(3^2 + 4^2)$ is going to be = 5. It is important to notice that the geometric concepts are easier to visualize in 2D or 3D; however, they are difficult to do. So, in higher dimensions, nonetheless since the fundamental mathematics remain the same what we can do is, we can understand these basic concepts using 2D and 3D geometry and then simply scale the number of dimensions, and then most of the things that we understand and learn will be the same at higher dimensions also.

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Vectors and distances

- Consider another example with two points X^1 and X^2

$$X^1 = \begin{bmatrix} x_1^1 \\ x_2^1 \end{bmatrix} \quad X^2 = \begin{bmatrix} x_1^2 \\ x_2^2 \end{bmatrix}$$

- The distance between these two can be calculated

$$l = \|X^2 - X^1\|_2$$

$$l = \sqrt{(x_1^2 - x_1^1)^2 + (x_2^2 - x_2^1)^2}$$

$$l = \sqrt{(X_2 - X_1)^T (X_2 - X_1)}$$

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So, in the previous slide we saw one point in 2 dimensions. Now, let us consider a case where we have 2 points in 2 dimensions. We have x_1 here, which has 2 numbers representing the 2 coordinates and we have x_2 here, which also represents the 2 coordinates. Now, we ask the question as to, whether we can define a vector which goes from x_1 to x_2 . So, pictorially this is the way in which, we are going to define this vector. What we do is, we draw a line starting from x_1 to x_2 and this vector is $x_2 - x_1$, the direction of the vector is given by this here, much like the previous case every vector will have a direction and a magnitude.

So, we might ask what is the magnitude of this vector and that is given by the wellknown formula that we see right here. Where what you do basically is, you take the x_1 coordinate of this point and this point take the difference square it, take the x_2 coordinate of this point the x_2 coordinate of this point, take the difference and square it add both of them and take a root and that is the equation that we have here.

This is the length of this vector right here, this also can be written in a compact form as given here, which is root of $(x_2 - x_1)^T (x_2 - x_1)$.

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Vectors and distances: Example

- What is the distance between points A and B , where A is $(2,7)$ and B is $(5,3)$
- Using the concept of distance introduced before

$$A = \begin{bmatrix} 2 \\ 7 \end{bmatrix}, B = \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$l = \sqrt{(5-2)^2 + (3-7)^2}$$

$$l = 5 \text{ units}$$

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Two simple examples to illustrate this, if I have 2 points A and B where A is 2 7 b is 5 3 then, the distances you take the difference between 5 and 2 and then square it and then, take the difference between 3 and 7 and then square it and then you will get your length as 5. So, that would be the length of the line that is, drawn between the 2 points A and B.

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Data science for Engineers

Unit vector

- A unit vector is a vector with magnitude 1 (distance from origin)
- Unit vectors are used to define directions in a coordinate system
- Any vector can be written as a product of a unit vector and a scalar magnitude

$$A = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \quad \checkmark \quad A = 5\hat{a}$$

Magnitude of A: $|A| = \sqrt{3^2 + 4^2} = 5 \checkmark$

$$\hat{a} = \frac{A}{|A|} = \begin{bmatrix} 3/5 \\ 4/5 \end{bmatrix}$$

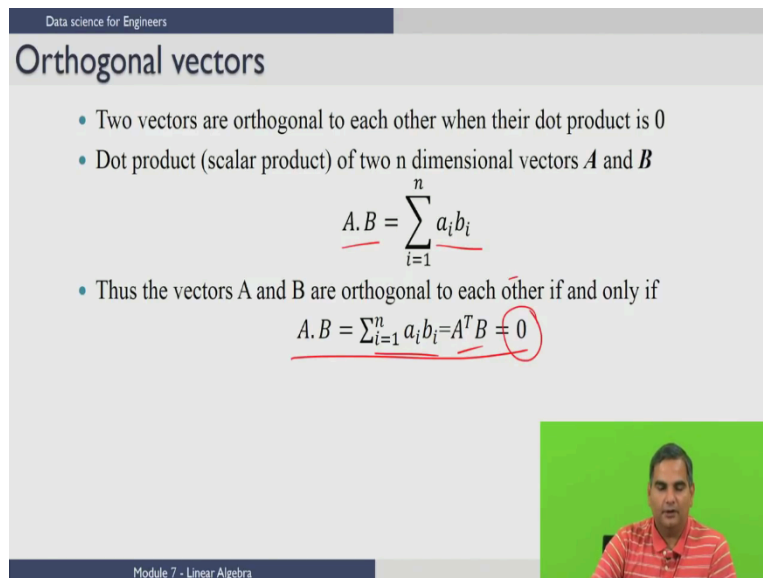
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Now, it is useful to define vectors with unit length, because once you write a vector in unit length any other vector in that direction, can be simply written as the unit vector times the magnitude of the vector that you are interested in.

So, how do I define a unit vector, let us take this vector $\begin{bmatrix} 3 \\ 4 \end{bmatrix}$, we know that the distance from the origin for this vector is root of $(3^2 + 4^2) = 5$. So, to define a unit vector what you do is, you take the vector and divide it by the magnitude of the vector. So, in this case it is 5. So, the unit vector becomes $\frac{3}{5}$ by $\frac{4}{5}$. So, the interesting thing is that, this unit vector is in the same direction as $\mathbf{a}$; however, it has magnitude 1. So, I could write $\mathbf{A}$ itself as 5 times $\mathbf{A}$. So now what has happened is this is a unit vector and this $\mathbf{a}$ has magnitude 5, which is what we derived here.

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Data science for Engineers

Orthogonal vectors

- Two vectors are orthogonal to each other when their dot product is 0
- Dot product (scalar product) of two n dimensional vectors $\mathbf{A}$ and $\mathbf{B}$

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^n a_i b_i$$
- Thus the vectors $\mathbf{A}$ and $\mathbf{B}$ are orthogonal to each other if and only if

$$\mathbf{A} \cdot \mathbf{B} = \sum_{i=1}^n a_i b_i = \mathbf{A}^T \mathbf{B} = 0$$

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We introduce the next concept, which is important for us to understand many of the things that we are going to teach. If there are 2 vectors, we call these vectors as orthogonal to each other when their dot product is 0. So, how do I define the dot product? So, if I take 2 vectors $\mathbf{A}$ and $\mathbf{B}$ $\mathbf{A} \cdot \mathbf{B}$ is simply $\sum_{i=1}^n a_i b_i$.

So, basically what you do is, if you have 2-dimensional vector then you take the 2 x coordinates multiply them, and then you take the 2 y coordinates and multiply them and add both of them you will get the dot product. This dot product again much like the distance that we saw before, can also be written in a compact form as $\mathbf{A}^T \mathbf{B}$ you can quite easily see that this and this will be the same, and if this dot product

turns out to be 0 then we call these vectors A and B as being orthogonal to each other.

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Orthogonal vectors: Example

- Consider the vectors v_1 and v_2 in 3D space. Identify if they are orthogonal to each other

$$v_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}$$

✓

$$v_2 = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$$

R Code
`v1=c(1,-2,4)`
`v2=c(2,5,2)`
`N=t(v1)%*%v2`

Console Output
`> N`
`[1]`
`[1,] 0`

- Taking the dot product of the vectors

$$\underline{v_1 \cdot v_2} = \underline{v_1^T v_2} = \underline{[1 \ -2 \ 4] \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}} = 0$$

- Hence, the vectors are orthogonal

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So, let us take an example to understand this, let us take 2 vectors in 3-dimensional space. Let us say, I have one vector which is 1 - 2 4 and I have the other vector which is 2 5 2 and if I take a dot product between these 2, which is $v_1^T v_2$ or $v_2^T v_1$, both will be the same. I have v_1^T which is 1 - 2 4 and this is 2 5 2, if this will be one times 2 - 5 times 2 + 4 times 2 you will see that goes to 0. So, we say that these 2 vectors are orthogonal to each other.

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Orthonormal vectors

- Orthonormal vectors are orthogonal vectors with unit magnitude
- Example

$$v_1 = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix} / \sqrt{1^2 + (-2)^2 + 4^2}$$

$$v_2 = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix} / \sqrt{2^2 + 5^2 + 2^2}$$

- Note that we have taken the vectors from the previous example and converted them into unit vectors by dividing them with their magnitudes.
- All orthonormal vectors are orthogonal

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Now, take the same 2 vectors, which are orthogonal to each other and you know that, when I take a dot product between these 2 vectors it is going to go to 0. If I also impose the condition, that I want each of these vectors to have unit magnitude then what I could possibly do? Is I could take this vector and then divide this vector by the magnitude of this vector.

So, this is going to be root of one squared + - 2 whole squared + 4 squared. Similarly, I can take this vector and divide this vector by the magnitude of the same vector, which is going to be root of 2 squared + 5 squared + 2 squared. Now, these 2 are unit vectors, because the magnitudes are the same and these unit vectors also turn out to be orthogonal to each other, the orthogonal property is not going to be lost, because these are scalar constants. So, while you take $v_1^T v_2$ or $v_2^T v_1$, it will still turn out to be 0. So, these vectors will still be orthogonal to each other. However now individually, they also have unit magnitude such vectors are called are orthonormal vectors, that we have defined here. Notice that all orthonormal vectors are orthogonal by definition.

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Data science for Engineers

Basis vectors

R^2

Let us consider two vectors $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 1 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 2v_1 + 1v_2$$

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 4 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 4v_1 + 4v_2$$

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 1v_1 + 3v_2$$

v_1 and v_2 are the basis vectors for R^2

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Now, we are going to come to the next interesting concept that we would need in data science quite a bit and I am going to explain this concept through very, very simple examples. This can also be very formally defined, what I am going to do is, I am going to try and explain this in a very simple fashion. So that you understand what this means and I also want to give a context, in terms of why these are some things that we are interested in looking at from a data science viewpoint.

So, we are going to introduce the notion of basis vectors. So, the idea here is the following, let us take R^2 which basically means that, we are looking at vectors in 2 dimensions. So, I could come up with many many vectors, right? So, there will be infinite number of vectors, which will be in 2 dimensions. So, this is like saying, if I take a 2-dimensional space how many points can I get? So, I can get infinite number of points. Which is what has been represented here.

So, I have put in some vectors and then these dots represent that, there are infinite number of such vectors in this space. Now, we might be interested in understanding, something more general than just saying that there are infinite number of vectors here. So, what we are interested in is, if we can represent all of these vectors using some basic elements and then some combination of these basic elements, is what we are interested in.

Now, let us consider 2 vectors for example, $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Now, if you take any vector that I have here, let us say take $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$, I can write $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ as some linear combination, of this vector + this vector. Similarly,

take $\begin{bmatrix} 4 \\ 4 \end{bmatrix}$, I can write $\begin{bmatrix} 4 \\ 4 \end{bmatrix}$ as a linear combination of this vector + this vector and that would be true for any vector that you have in this space.

So, in some sense what we say is that, these 2 vectors characterize the space or they form a basis for the space and any vector in this space can simply be written as a linear combination of these 2 vectors. Now you notice, the linear combinations are actually the numbers themselves. So, for example, if I want this to be written as a linear combination of $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, the linear combination the scalar multiples are 2 which is this, and 1 which is this similarly 4 here 4 here and so on.

So, the key point being, while we have infinite number of vectors here, they can all be generated as a linear combination of just 2 vectors and we have shown here, these 2 vectors as $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Now, these 2 vectors are called the basis for the whole space, if I can write every vector in the space as a linear combination of these vectors and these vectors are independent of each of them.

Then we call them as a basis for the space. So, why do you want these vectors to be independent of each other? We want these vectors to be independent of each other, because we want every vector, that is in the basis to generate unique information. If they become dependent on each other, then this vector is not going to bring in anything unique. So, basis has 2 properties, every vector in the basis should bring something unique, and these vectors in the basis should be enough, to characterize the whole space, in other words the vector should be complete.

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Data science for Engineers
Basis vectors

- Basis vectors are set of vectors that are independent and span the space
- Example:
 - Two vectors $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
 - Can span R^2 and are independent and hence form the basis for the R^2 space.



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So, this we can formally say as the following, basis vectors for any space are a set of vectors that are independent and span the space and the word span basically means that, any vector in that space, I can write as a linear combination of the basis vectors. So, the previous example, we saw that the 2 vectors $v_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $v_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, can span the whole $\mathbb{R}^2$ and you can clearly see that they are independent of each other, because no multiple scalar multiple of this will be able to give you this vector .

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Basis vectors are not unique

Consider two vectors $v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} = 1.5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-0.5) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1.5v_1 + (-0.5)v_2$$

$$\begin{bmatrix} 4 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 0 \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 4v_1 + 0v_2$$

$$\begin{bmatrix} 1 \\ 3 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-1) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 2v_1 + (-1)v_2$$

Hence, this v_1 and v_2 are also basis vectors for $\mathbb{R}^2$

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So, the next question that immediately pops up in ones head is, if I have a basis vector, are they unique? Now it turns out these basis vectors are not unique, you can find many many sets of a basis vectors, all of which would be equivalent. The only conditions are that they have to be independent and should span the space. So, take the same example and let us consider 2 other vectors, which are independent.

So, the same example as before, where we had used 2 basis vectors $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$, I am going to replace them by $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Now, the first thing that we have to check is, if these vectors are linearly independent or not and that is very easy to verify. If I multiply this vector by any scalar, I will never be able to get this vector. So, for example, if I multiply this by -1 I will get -1 and -1, but not $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. So, these 2 are linearly independent of each other.

Now, let us take the same vectors and then see what happens. So, remember we represented $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ in the previous case, as 2 times $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ + 1 times $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Now, let us see whether I can represent this $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ as a linear combination of $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$. So, if you look at this, this is the linear combination notice; however, because of the way I have chosen these vectors, these numbers are not the same as this.

So, in the previous case when we use $1\ 0$ on $0\ 1$, we said this can be written as 2 times $1\ 0 + 1$ times $0\ 1$; however, the numbers have changed now, nonetheless I can write this as a linear combination of these 2 basis vectors.

Let us take this $4\ 4$ as an example. So, that can be written as an interesting linear combination, which is 4 times $1\ 1 + 0$ times $1 - 1$ right? So, that will give you $4\ 4$ similarly $1\ 3$ can be written as, 2 times $1\ 1 + -1$ times $1 - 1$. So, this is another linear combination of the same basis vectors.

So, the key point that I want to make here is that, the basis vectors are not unique there are many ways in which you can define the basis vectors; however, they all share the same property that, if I have a set of vectors which I call as a basis vector, those vectors have to be independent of each other and they should span the whole space and whether you take $0\ 1\ 1\ 0$ and call it a basis set or you take $1\ 1$ and $1 - 1$ and call the basis set, both are all right and you can see that, in each case the vectors are independent of each other and they span the whole space.

An interesting thing to note here though is that, I cannot have 2 basis sets which have different number of vectors, what I mean here is in the previous example though the basis vectors were $1\ 0$ and $0\ 1$, there were only 2 vectors. Similarly, in this case the basis vectors are $1\ 1$ and $1 - 1$.

However, there are still only 2 vectors. So, while you could have many sets of basis vectors, all of them being equivalent, the number of vectors in each set will be the same. They cannot be different and this is easy to see. I am not going to formally show this, but this is something that you should keep in mind, in other words for the same space you cannot have 2 basis sets - one with n , vectors other one with m vectors - that is not possible. So, if it is a basis set for the same space, the number of vectors in each set should be the same. Now, I do not want you to think that the basis set will always have to be the number of elements in the vector.

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Data science for Engineers

Basis vectors

Consider two vectors $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ and $v_2 = \begin{bmatrix} 4 \\ 1 \\ 2 \\ 3 \end{bmatrix}$

$\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} = 1 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + 0 \begin{bmatrix} 4 \\ 1 \\ 2 \\ 3 \end{bmatrix} = 1v_1 + 0v_2$

$\begin{bmatrix} 7 \\ 7 \\ 11 \\ 15 \end{bmatrix} = 3 \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix} + 1 \begin{bmatrix} 4 \\ 1 \\ 2 \\ 3 \end{bmatrix} = 3v_1 + 1v_2$

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So, to give you another example, we have generated this data in a particular fashion. Consider now this set of vectors right? There are infinite number of vectors here and we will say all of these vectors are in space $\mathbb{R}^4$, which basically means that there are 4 components in each of these vectors.

Now, what we want to ask is, what is the basis set for these kinds of vectors? Now when I do this here, the assumption is the extra vectors that I keep generating, the infinite number of them, all follow certain pattern that these vectors are also following and we will see what that pattern is. So, what we can do is, we can take, let us say 2 vectors here, in this case this is how this example has been constructed to illustrate an important idea. Let us take this vectors v_1 which is $\begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}$ and let us take some vector here, in this set let us take this vector here, and then see what happens, when I try to write it as a linear combination of these 2 vectors.

So, I can see that if I take this I can write it as 1 times this + 0 times the second vector. So, that is one linear combination, now let us take some other vector here. So, let us say for example, we have taken this vector $\begin{bmatrix} 7 \\ 7 \\ 11 \\ 15 \end{bmatrix}$, we can see that that can be written as a linear combination of 3 times the first vector + 1 times the second vector and so on.

Now, you could do this exercise for each one of these vectors and you will be able to see, because of the way we have constructed these vectors, you will be able to see that each one of these vectors, I can write as a linear combination of v_1 and v_2 . So, what this basically says

is the following, it says that, though I have 4 components in each of these vectors, that is, all of these vectors are in $\mathbb{R}^4$, because of the way in which these vectors have been generated, they do not need 4 basis vectors to explain them, all of these vectors have been derived as a linear combination of just 2 basis vectors, which are given here and here.

So, in other words all of these vectors would occupy A_2 dimensional, what we call as a subspace in $\mathbb{R}^4$ right? So, if you take every vector in $\mathbb{R}^4$, without leaving out anything then, you would need 4 basis vectors to explain all of them. However, these vectors have been picked in such a way, that they are only linear combination of these 2 vectors. So, I just need 2 vectors to represent all of this. So, I say that, all of these vectors fall in a 2 dimensional subspace in $\mathbb{R}^4$.

So, this is an important concept of subspace, which is very, very important for us from a data science viewpoint and I am going to explain to you why. We are interested in things like this, from a data science viewpoint. Now, the next question that we might ask is the following.

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Finding basis vectors

$$\begin{bmatrix} 6 \\ 5 \\ 8 \\ 11 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 9 \\ 4 \\ 7 \\ 10 \end{bmatrix}, \begin{bmatrix} -3 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ -1 \\ -1 \\ -1 \end{bmatrix},$$

$$\begin{bmatrix} 14 \\ 7 \\ 12 \\ 17 \end{bmatrix}, \begin{bmatrix} 11 \\ 8 \\ 13 \\ 18 \end{bmatrix}, \begin{bmatrix} 7 \\ 0 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ -3 \\ -4 \\ -5 \end{bmatrix}, \begin{bmatrix} 7 \\ 7 \\ 11 \\ 15 \end{bmatrix}$$

2D

Evaluate the rank of the matrix

$$\begin{bmatrix} 6 & 1 & 9 & -3 & 3 & 14 & 11 & 7 & 2 & 7 \\ 5 & 2 & 4 & 1 & -1 & 7 & 8 & 0 & -3 & 7 \\ 8 & 3 & 7 & 1 & -1 & 12 & 13 & 1 & -4 & 11 \\ 11 & 4 & 10 & 1 & -1 & 17 & 18 & 2 & -5 & 15 \end{bmatrix}$$

Rank of the matrix is: 2

Any two independent columns can be picked from the above matrix as basis vectors

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So, this is the same as the previous slide, except that I have removed the dot dot dot. So, the way to think about this it is let us say there is some data generation process, which is generating vectors like this, and the dot dot dots that I have left out, will also be generated in the same fashion, because those are also vectors that are being generated by the same data generation process.

So, I have certain data generation process and I am generating samples from that and I have done let us say 10 experiments. So, I have got these 10 samples and the other dots will be similar, now what I want to know is if you give me these, vectors in $\mathbb{R}^4$, how many basis vectors do I need to represent them? In the previous slide I had already shown you what the basis vectors are and then shown how I could generate many many linear combinations of just 2 in $\mathbb{R}^4$ to get a subspace. I am looking at an inverse problem here, where I do not know what are the vectors that are generating these samples, nonetheless I have got enough samples.

Let us say 10 and if I were to continue this experiment and if it was the same data generation process, I might get 20 samples 30 samples and so on; however, what I want to know is with these 10 samples, how do I find the basis vectors? So, we are going to use concepts that we have learned before to do this. If we were to stack all of these vectors in a matrix like this.

So, this is a first vector here, from here second vector and so on all the way up to the last vector and I say I have so many vectors, how many fundamental vectors do I need to represent all of these as linear combinations? It is a question that I am asking. The answer is straightforward this is something that we have already seen before, if you identify the rank of this matrix it will give you the number of linearly independent columns.

So, what that basically means is, if I get a certain rank for this matrix, then it tells me there are only so many linearly independent columns and every other column, can be written as a linear combination of those independent columns. So, while I have many many columns here, 1 2 all the way up to 10. The rank of the matrix will tell me, how many are fundamental to explaining all of these columns, and how many columns do I need.

So that I can generate the remaining columns as a linear combination of these columns, and as I have been mentioning again, if the data generation process remains the same as I add more and more columns to these, they will also be linear combinations of the columns that we identify here. So, when we go ahead and try to find the rank of this matrix, the rank of the matrix will turn out to be 2 and it will turn out to be 2 because, of the way we have generated this data.

Now, if you had generated these vectors in such a way that they are a linear combination of 3 vectors, then the rank of the matrix would have been 3. If you had generated these vectors in such a manner, that they are linear combinations of 4 linearly independent vectors, then the rank of the matrix would have been 4, but that would be the maximum

rank of the matrix, because in $\mathbb{R}^4$ you would not need more than 4 linearly independent vectors to represent all the vectors.

So, the maximum rank can be 4, the rank could be 1 2 or 3. If it is 1 then I have only 1 basis vector, if there are 2 there are 2 basis vectors 3 there are 3 basis vectors and so on. In this case since the rank of the matrix turns out to be 2, there are only 2 column vectors that I need to represent every column in this matrix. So, the basis set has size 2, is something that we have determined. The next question is the basis set is size 2, what are the actual vectors? What we can do is, we can pick any 2 linearly independent columns here and then those could be the basis vectors.

So, for example, I could choose this and this and say, this is the basis vector for all of these columns or I could choose this and this and this or this and this and so on. So, I can choose any 2 columns, as long as they are linearly independent of each other and this is something that we know, from what we have learned before, because we already know that the basis vectors need not be unique. So, I pick any 2 linearly independent columns that represents this data. Now, let me take a minute to explain why this is important from a data science viewpoint. I will just show you some numbers. Supposing, I have let us say 200 such samples and I want to store these 200 samples since each sample has 4 numbers, I would be storing 200 times 4 which is 800 numbers.

Now, let us assume we do the same exercise for these 200 samples and then we find that, we have only 2 basis vectors, which are going to be 2 vectors out of this set. What I could do is, I could store these 2 basis vectors that, would be 8 numbers which is 2 by 4 and for the remaining 198 samples, instead of storing all the samples and all the numbers in each of these samples, what I could do is for each sample I could just store 2 numbers right?

So, for example, if you take this sample, instead of storing all the 4 numbers, I could just store 2 numbers, which are the linear combinations that I am going to use to construct this. So, for example, since I have 2 basis vectors here, there is going to be some number α_1 times the basis vector, + α_2 times the second basis vector, which will give me this sample right?

So, instead of storing these 4 numbers, I could simply store these 2 constants and since I already have stored the basis vectors, whenever I want to reconstruct this, I can simply take the first constant and multiply v_1 + the second constant multiply v_2 and I will get this number. So, I store 2 basis vectors which gives me 8 numbers and then for the remaining 198 samples, I simply store 2 constants. So, this would give me $396 + 8 = 404$ numbers stored. I will be able to reconstruct the whole data set.

So, compare that with 800. So, I have half reduction in number. So, when you have vectors in multiple dimensions, let us say you have vectors in 10 dimensions 20 dimensions and the number of basis vectors, are much lower than those numbers. So, for example, if you have 10-dimensional vector and the basis vectors are just 3, then you can see the kind of reduction that you will get in terms of data storage. So, this is one viewpoint from data science. Why? It is very important to understand and characterize the data in terms of what fundamentally characterizes the data. So that you can store less, we can do smarter computations and there are many other reasons why we will want to do this, you can identify this basis to identify a model between this data, you can identify a basis to do noise reduction in the data and so on.

So, all of those viewpoints we will talk about as we go forward, with this data science course. In the next lecture, we will continue and then try and understand how we can use these concepts. The notion of basis vectors, the notion of orthogonality to understand concepts such as projections, hyper planes, half spaces and so on, which all are critical from a data science viewpoint. So, I will pick up from here in the next lecture

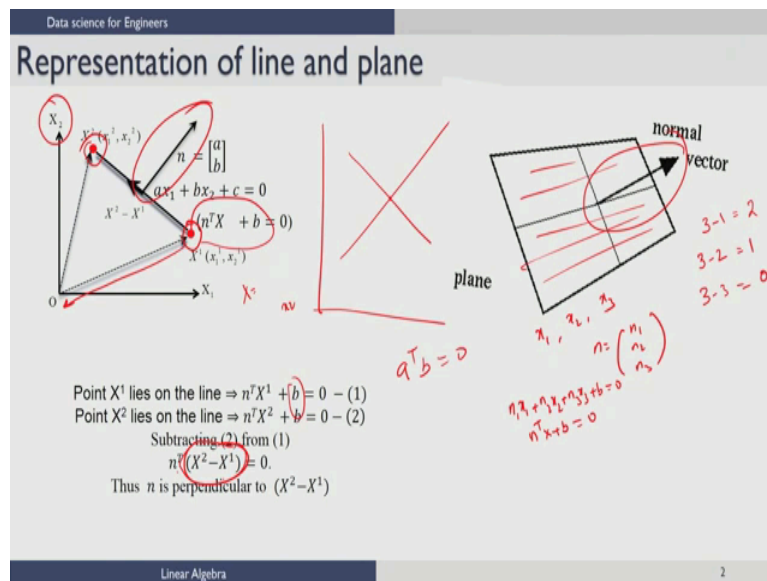
Thank you.

Data Science for Engineers
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Lecture - 16
Linear Algebra - Distance, Hyperplanes and Halfspaces, Eigenvalues, Eigenvectors
(Continued 1)

So, let us continue with our lectures on linear algebra for data science. We will continue discussing distances hyperplanes, half spaces, eigenvalues and eigenvectors in this lecture, and the lecture that follows this lecture. So, what we are going to do is we are going to think about the equations in multi-dimensional space. And then think about what geometric objects that these equations represent.

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So, let us look at what equations mean from a geometric viewpoint. To do this, let us start with 2 dimensions. Let us assume that we have a 2 dimensional space in X_1 and X_2 , and let us also assume that we have one equation that relates the variables; X_1 and X_2 which is $aX_1 + bx + c = 0$.

So, we want to understand what geometry this equation represents. It turns out in this case of the 2-dimensional space, this equation represents a line which is depicted here. So, a single equation in a

2-dimensional space represents a line. So, one might ask what does 2 equations represent.

And to understand this if you look at a picture like this, let us say I have one equation which is a line, let us draw the other equation which is also a line. Then if both of these equations have to be satisfied, then that has to be this intersecting point. So, 2 equations in 2 variables represent a point if these equations are solvable together.

Now, if you have no relationships between these variables, then we would say that we are representing all the points in the 2-dimensional space. And there is no relationship that constrains these points to either lie on a line or be a single point and so on. Now let us look at this equation, and then rewrite it in a form that is generally used. And that form is the following which is $n^T X + b = 0$, n is this column vector that is defined here. X is a vector of variables X_1 and X_2 . And if you do a one to one comparison between this equation and this equation, you will see that $b = c$. So, a general equation can be written in this form $n^T X + b = 0$.

Now, we want to understand what this n depicts in this example. Now if you look at this picture here, we have shown n as a normal to this line. And let us see why that is true. To see that, let us first start by looking at 2 points on the line. So, let us start with this point X_1 and then X_2 . Notice that both these points are on the line.

So, when I substitute X_1 into the equation for the line, it should satisfy it which is what is shown here $n^T X_1 + b = 0$ and when I substitute X_2 into the line equation that should also satisfy. So, $n^T X_2 + b = 0$. Now what you could do is you could subtract the first equation from the second equation. The b 's will get cancelled. You will have $n^T X_2 - X_1 = 0$. Now let us interpret this equation. From vector addition you know that if I have X_1 , $-X_1$ is in this direction.

So, when I do $X_2 - X_1$ I am adding X_2 1 - X_1 , which is equivalent to starting from here and going here ; which is basically through vector addition in the direction of this line. So, what this equation basically tells us is that this is in the direction of the line.

And from our orthogonality lecture we saw before, we said if $A^T b = 0$, then a and b are orthogonal. Since n^T this quantity is 0, and this quantity is in the direction of the line n has to be perpendicular to the line. So, this is a very important idea that we will use in data science quite a bit, when we looked at linearly separable classes. And then classifying classes that are linearly separable.

Now, if you want to extend this, and then ask the question, if I have one equation in a 3-dimensional space, what does that represent. Now the form of the equation will be very similar. You will have something

like this here, and n now would become supposing you have 3 variables $X_1 X_2 X_3$. N could be $n_1 n_2 n_3$.

So, the same equation would be $n_1 X_1 + n_2 X_2 + n_3 X_3 + b = 0$. So, which is $n^T X + b = 0$. So, irrespective of what-ever is the dimension of your system, you can always represent a single linear equation in this form $n^T X + b = 0$.

Now, we ask the question as to what a single equation would represent in a 3 - dimensional space. And in a 3-dimensional space a single equation would represent a plane a 2-dimensional object. The way to see this is in 3 dimensions we have 3 degrees of freedom. If you write one equation you are taking away one degree of freedom. So, we are left with 2 degrees of freedom.

And a 2 degree of freedom object is basically a plane. And this is what is shown here. So, if I have one equation, that equation itself would represent this plane right here; which is what we see. And very similar to how we drew the normal to the line here, this n would represent a normal to the plane. So, this would represent a projection outside out of the plane orthogonal to the plane. So, that is what n would represent.

So, in 3 dimensions, one equation would represent a plane. And in 3 dimensions 2 equations would represent a line, because we have 3 degrees of freedom if you take away 2, then you have one, a one-dimensional object is a line. And if you have 3 equations, then you have $3 - 3 = 0$, the 0-dimensional object would be a point, and this would be a point as long as these 3 equations are consistent and solvable.

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Data science for Engineers

Projections

- We can define the projection ($\hat{X}$) of a vector (X) onto a lower dimension (two dimensions in the picture) mathematically as

$$\hat{X} = c_1 v_1 + c_2 v_2$$
- Using vector addition

$$X = c_1 v_1 + c_2 v_2 + n$$

Handwritten notes: v_1, v_2 (circled), $n^T v_1 = v_1^T n = 0$, $n^T v_2 = v_2^T n = 0$

Linear Algebra

Now, that we have talked about what equations represent and so on. One of the things that we are quite interested in and you will see this again and again in data science, as we teach some of the algorithms later such as principle component analysis and so on. We are always interested in projecting vectors onto surfaces. The reason why we are interested in doing this is, because many times we might want to represent data through a smaller set of objects or a smaller number of vectors. So, in some sense the data cannot be completely represented by these vectors.

So, we might ask the question as to what is the best approximation for this data point based on the vectors that I want to represent this data point with. So, this is a very important question that we will keep asking again and again. You will understand this in much more detail and clarity, once we talk about some of the data science concepts.

For now, I am just going to treat this mathematically, and then explain to you how we do projections and what are the equations that we can get for writing down projections. The interpretation for this and the use for this in data science is something that we will see as we go along this course later. So, let us take a very, very simple example. Let us assume that I have a plane which is shown here in this picture.

Since I have a plane basically we are looking at A_3 -dimensional space. A plane has to be represented by 2 dimensions, because it is a 2-dimensional object. Let us assume that the basis vectors for this plane are v_1 and v_2 . We have already discussed what basis vectors are in a previous lecture. So, for a 2-dimensional object we will need 2 basis vectors.

So, let us assume these basis vectors are v_1 and v_2 . Just to recap what this basis vectors are useful for is that, any line on this plane, basically can be written as a linear combination of v_1 and v_2 . That is what we described before, that these basis vectors are enough to characterize every point or any vector on this 2-dimensional plane. So, any vector can be written as a linear combination of v_1 and v_2 .

Now, the way this picture is drawn, you would see that this is the plane and I have let us say, a vector that is coming out of the plane. So, this is not clearly in the plane. So, it is projecting out. So, from the data science viewpoint if you want to make an analogy, what we are saying here is that, I have a data here X which is represented by this vector. I want to write this simply as only a function of v_1 and v_2 . So, in other words I want to represent this vector X , in a tool, I cannot do it exactly projecting out of the plane. So, I might ask what is the next best thing that I could do in this case. It turns out the next best thing to do would

be to project this vector onto the plane, because ultimately, however I write this vector with only this 2 basis vectors it has to be on the plane.

Now, there are many vectors on the plane. I want to find what is the best projection for this onto this plane. So, a common sense idea would be to say, I want a point here which I write. And if this is the projection of this vector I want this distance to be minimized. So, you can see why that is. Think about this if you keep projecting it back to the plane, if this is the closest point if the vector is already in the plane, it would be the same vector that is also the product right. So, as soon as this vector goes up slightly outside the plane, I want it to be projected back. So, that it is closest to that point of projection. So, how do we explain these concepts mathematically? So, we do that here. First, $\hat{X}$ is the projection of X onto lower dimension in this case 2 dimensions.

And since $\hat{X}$ has to be in the lower dimension, We already know that it can be written as a linear combination of v_1 and v_2 . So, $\hat{X}$ is $c_1 v_1 + c_2 v_2$. The c_1 and c_2 are yet to be determined. So, we do not know what those are. We are going to try and determine these 2 using this idea of projection. So, what we are going to say is that, if this is the projection, then the closest point from here would be when I draw a perpendicular or drop a perpendicular onto the plane. So, as long as these 2 points when I connect by vector n , that vector is perpendicular to this plane, then I would have found the closest point on this plane, which is what I am going for in terms of projections.

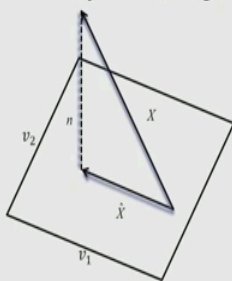
So, using vector addition, again we can start from here let us say, and this is x . So, X can be written as $\hat{X} + n$, which is what is written here. And $\hat{X}$ has been expanded to be $c_1 v_1 + c_2 v_2$. Notice that while we write this, the fact that we are using a projection comes from this n being perpendicular to the plane. So, what does n being perpendicular to the plane mean? If n is perpendicular to the plane, then we know that $n^T v_1$ or $v_1^T n$ both are the same will be 0. Similarly, $n^T v_2 = v_2^T n$ will also be 0. So, these are 2 facts that will know, if n is perpendicular to the plane. So, how are we going to use this to calculate c_1 and c_2 is what I am going to show you in the next slide.

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Data science for Engineers

Projections

- Projections onto general orthogonal directions (two dimensions in this case)



$$v_1^T n = 0$$

$$v_1^T (X - c_1 v_1 + c_2 v_2) = 0$$

$$v_1^T X - c_1 v_1^T v_1 = 0$$

$$\hat{X} = \frac{v_1^T X}{v_1^T v_1} v_1 + \frac{v_2^T X}{v_2^T v_2} v_2$$

Handwritten notes in red ink show the derivation of the projection coefficients:

$$c_1 = \frac{v_1^T X}{v_1^T v_1}$$

$$c_2 = \frac{v_2^T X}{v_2^T v_2}$$

Linear Algebra

So, let us first take this $v_1^T n = 0$; the first equation I wrote. Let me write n as this from the previous slide, because X was $c_1 v_1 + c_2 v_2 + n$, I simply move $c_1 v_1$ and $c_2 v_2$ to the other side. And I have this equation right here.

Now, when I expand this equation, I will get $v_1^T X - c_1 v_1^T v_1$ and I will also have another term which would be here $-c_2 v_1^T v_2$. Now as the first case I am going to show you how you do projections on 2 orthogonal directions. Now if these 2 directions are orthogonal the basis vectors themselves are orthogonal, then we know that this will be 0, that is the reason why this term drops out. And I have $v_1^T X - c_1 v_1^T v_1 = 0$. Take this to the other side, and then bring $v_1^T v_1$ to the denominator, then you will get $c_1 = v_1^T X$ divided by $v_1^T v_1$.

Now, you could use the same idea, and then do the calculations for $v_2^T n = 0$. And when you do this, again you use this fact that $v_2^T v_1$ or $v_1^T v_2 = 0$ because these are orthogonal directions. And then you will end up with this equation for c_2 , which will be $v_2^T X$ divided by $v_2^T v_2$.

Once you get this, then you can back out the projection and the projection is c_1 times v_1 + c_2 times v_2 . So, this is how you project a vector on to 2 orthogonal directions, and this can be extended to 3 orthogonal directions 4 orthogonal directions and so on. Because all you will get if let us say it is 3 orthogonal directions then you will get $v_1^T X$ divided by $v_1^T v_1$ for c_1 , this is for c_2 and $v_3^T X$ divided by $v_3^T v_3$ for the third constant c_3 .

So, this is how you do projection. This is a very, very important idea, and this will be used in many many places in data science. So, it is worthwhile to clearly understand this.

(Refer Slide Time: 17:52)

Data science for Engineers

Projections: Example

$$X = [1 \ 2 \ 3]^T$$

- Projecting this vector onto the space spanned by the vectors $v_1 = [1 \ -1 \ -2]^T$ and $v_2 = [2 \ 0 \ 1]^T$
- Thus, finding the projection onto the plane defined by v_1 and v_2 is

$$\hat{X} = \frac{[1 \ 2 \ 3] \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix}}{6} \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} + \frac{[1 \ 2 \ 3] \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}}{5} \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

Linear Algebra 5

Now, let us move on to doing an example for this projection. Let us take a very simple example, let us say I have vector 1 2 3 transpose; so, column vector. So, this is a vector in a 3-dimensional space.

Now, let us take 2 vectors and then make a plane. So, let us take vector v_1 which is 1 - 1 - 2, and v_2 which is 2 0 1. And then try and see whether I can project X on to these. Let us first find out whether these 2 vectors are orthogonal. So, to do that we have to do $v_1^T v_2$. So, I am going to do 1 - 1 - 2 2 0 1. So, this will be one times 2 - 1 times 0 0 - 2 times 1 2. So, 2 - 2 is 0.

So, we know that these 2 vectors are orthogonal. So, we can use a formula that we had before. Now this formula is what we apply here. So, this is $v_1^T X$ transpose, sorry, this is $X^T v_1$ which is 1 - 1 - 2, and this should be $v_1^T v_1$. So, that will be one square + 1 square + 2 square. So, 1 + 1 + 4, 6. So, this is constant c_1 that we get, and this is multiplied by v_1 .

And if you look at the second term here. So, this is X^T this is v_2 2 0 1. And this should be $v_2^T v_2$. So, this should be $2^2 + 0^2 + 1^2 = 5$. And we have this vector v_2 .

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Data science for Engineers

Projections: Example

$$\hat{X} = \frac{-7}{6} \begin{bmatrix} 1 \\ -1 \\ -2 \end{bmatrix} + 1 \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}$$

$$\hat{X} = \begin{bmatrix} 5/6 \\ 7/6 \\ 20/6 \end{bmatrix}$$

Handwritten red annotations: A red circle around the vector $\begin{bmatrix} 5/6 \\ 7/6 \\ 20/6 \end{bmatrix}$ and a red arrow pointing from the vector $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ to the vector $\begin{bmatrix} 5/6 \\ 7/6 \\ 20/6 \end{bmatrix}$.

Linear Algebra



So, once you simplify this further you get the projection as the following. So, my original data vector 1 2 3, when it is projected onto a space spanned by these 2 basis vectors becomes this.

So, in other words, if I had a data point 1 2 3, and then I say I want to represent this with only 2 vectors that I had identified before whatever reason it might be, then the best representation is the following is what this projection says.

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Data science for Engineers

Projection -Generalization

- Projections onto general directions
- Consider the problem of projection of X onto a space spanned k linearly independent vectors

Handwritten red annotations: $X \in \mathbb{R}^n$, $v_1, \dots, v_k \in \mathbb{R}^n$

$$\hat{X} = \sum_{j=1}^k c_j v_j$$

Handwritten red annotations: $c_1 v_1 + c_2 v_2 + \dots + c_k v_k$

$$\hat{X} = \begin{bmatrix} v_1 & \dots & v_k \end{bmatrix} \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}$$

$$\hat{X} = Vc$$

Linear Algebra



Now, we talked about projecting on to certain number of directions. And we also talked about projections when these directions are orthogonal. I am going to generalize this in the coming slides. So, that we have a result that is general and can be used in many places. So, I am going to look at how we can project this vectors onto general directions. So, let us consider the problem of projection of X onto space spanned by k linearly independent vectors,. Now I have dropped this notion of orthogonal here, I am simply saying these vectors are independent. As before since I want to project X on to k linearly independent vectors.

I am going to represent that projection as $\hat{X}$. And because this $\hat{X}$ is in a space spanned by this k linearly independent vectors, I can write this as a linear combination of these k vectors; which is what I have written here. So, if you expand this you will get $c_1 v_1 + c_2 v_2$ and so on $+ c_k v_k$. Notice in this equation it is important to really understand this carefully. Notice in this equation $v_1 v_2 v_k$ are all vectors, and $c_1 c_2 c_k$ are scalar constants.

We can write this equation also in this form. Where, what we do is we stack these vectors, into a matrix. So for example, if X is in an n dimensional space R^n . Then we would assume that each of these vectors are also in R^n . So, v_1 to v_k are all element of R^n . And when you stack k vectors like this in a matrix, then you would get a matrix of dimension n by k .

And since there are k constants, which I have put in a vector. So, this would be a vector of dimension k by 1 . And you can notice that this n by k times k by 1 will give you an n by 1 vector which is what this $\hat{X}$ nonetheless, this n by one vector is in a space spanned by these k vectors linearly independent vectors.

Now this is an important thing to notice, if you go back and then say let me expand this, then basically you should get this. And this is another way of thinking about matrix multiplications, which is important to understand. So, let me illustrate this with some very, very simple examples so that we use this at later times.

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Data science for Engineers

Projection -Generalization

- Projections onto general directions
- Consider the problem of projection of X onto a space spanned k linearly independent vectors

$$\hat{X} = \sum_{j=1}^k c_j v_j$$

$$\hat{X} = [v_1 \dots v_k] \begin{bmatrix} c_1 \\ \vdots \\ c_k \end{bmatrix}$$

$$\hat{X} = Vc$$

$$= c_1 v_1 + c_2 v_2 + \dots + c_k v_k$$

Handwritten notes in red:

$X \in \mathbb{R}^n$
 $v_1, \dots, v_k \in \mathbb{R}^n$

Example calculation:
 $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+1 \\ 0+1 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$
 $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Linear Algebra 7

So, if I have let us say a matrix $0 \ 0 \ 0 \ 1$, I will say, and I want to multiply this by $0 \ 0$. The standard way of doing this would be one times one + 1 times one and 0 times 1 + 1 times 1.

So, this will be $2 \ 1$, right. This is your standard matrix multiplication that you have seen. You can also interpret this slightly differently. You could say that this matrix multiplication is also one times this vector $1 \ 0 + 1$ times this vector. So, this is what we can think of this, matrix multiplication as. Now if

you notice this will also give you the result $2 \ 0 \ 0$. So, what we are doing is there are many columns here and there are, these scalar constants much like this. So, when we multiply this this will be $c \ 1$ times the first column; much like, how we have written here. So, this will be $c \ 1$ times $v_1 + c_2$ times v_2 , all the way up to c_k times v_k .

So, this and this are same and you see that this is this. So, $\hat{X}$ can be written as v times c ; where v is a matrix where are all these basis vectors are stacked in columns. And c are the scalar constants which have been stacked as a single column. Now, let us proceed to identify the projection from here.

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Data science for Engineers

Projection -Generalization

- Using orthogonality idea

$X = \hat{X} + n$
 $n = X - \hat{X}$

$$V^T(X - \hat{X}) = V^T(X - Vc) = 0$$

$$\begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \end{bmatrix} V^T X - V^T V c = 0$$

$$c = (V^T V)^{-1} V^T X$$

$$\hat{X} = V(V^T V)^{-1} V^T X$$

Linear Algebra

We then use the orthogonality idea. Remember, we have $X = \hat{X} + n$. That means, $n = X - \hat{X}$ and if $\hat{X}$ has to be a projection, then n has to be a vector that is orthogonal to the space spanned by the k linearly independent vectors. For n to be orthogonal to a plane to a geometric object spanned by this k linearly independent vectors. n has to be orthogonal to every one of these vectors.

So, that is what we write here in a matrix form instead of writing $v_1^T X - \hat{X}$ is 0 $v_2^T X - \hat{X}$ is 0 and so on. So, we write this in a matrix form where we say v^T there I will have $v_1^T v_2^T$ transpose. All the way up to v^T times $X - \hat{X} = 0$.

Now, $\hat{X}$ from the previous slide was v times c . So, we have $v^T X - v^T c = 0$. So, if I expand this I will get $v^T X - v^T v c = 0$. If I take this term to the other side, and then do the inverse. I will get $c = v^T v$ inverse $v^T X$. Whenever we take inverses we have to always make sure that we can actually identify an inverse. In this case I will be guaranteed to have an inverse for $v^T v$ if the columns of v are linearly independent. And the fact that we have chosen these basis vectors as linearly independent already, assures us that those are linearly independent.

So, this inverse is something that exists. Once we calculate this c we know $\hat{X}$ is v times c . So, I simply plug this $v c$ back in and I get the expression for projection. So, this is how you do projection onto general directions. Now this is a very important idea that is used in several data science algorithms. In fact, this is a backbone for

something called principal component analysis. And this is also used in many many other machine learning algorithms.

So, it is important to understand this idea very clearly. Now that we have understood projections, in the next lecture I will describe the notion of an hyper plane and half spaces. And then continue on to eigenvalues and eigenvectors. I will see you the next lecture.

Thank you.

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Lecture- 17

Linear Algebra - Distance, Hyperplanes and Halfspaces, Eigenvalues, Eigenvectors

We will continue with our lectures on linear algebra for data science. Today I will talk about hyper planes, half spaces and eigenvalues, eigenvectors and so on.

(Refer Slide Time: 00:27)

Data science for Engineers
Hyperplanes

- Geometrically, hyperplane is a geometric entity whose dimension is one less than that of its ambient space.
- For instance, the hyperplanes for a 3D space are 2D planes and hyperplanes for a 2D space are 1D lines and so on.
- The hyperplane is usually described by an equation as follows

$$X^T n + b = 0$$

$x_1 n_1 + x_2 n_2 + x_3 n_3 + \dots + x_n n_n + b = 0$
 $x_1 n_1 + x_2 n_2 + b = 0$
 $X^T n = 0$

Linear Algebra
2

Let us start this lecture with hyper planes. Geometrically hyper plane is a geometric entity whose dimension is 1 less than that of its ambient space. So, what this means is, the following. For example, if you take the 3D space then hyper plane is a geometric entity which is 1 dimension less. So, its going to be 2 dimensions and a 2 dimensional entity in a 3D space would be a plane.

Now if you take 2 dimensions, then 1 dimension less would be a single dimensional geometric entity, which would be a line and so on. The hyper plane is usually described by an equation as follows, if I expand this out for n variables. So, I will get something like $X_1 n_1 + X_2 n_2 + X_3 n_3$ and so on $X_n n_n + b = 0$ in just two dimensions, you will see that this is $X_1 n_1 + X_2 n_2 + b = 0$ which is an equation of line. We have

seen before the idea of subspaces. Hyperplanes in general are not subspaces, however, if we have hyper planes of the form $X^T n = 0$; that is if the plane goes through the origin, then an hyper plane also becomes a subspace.

(Refer Slide Time: 02:03)

Data science for Engineers

Halfspace

- We can observe that the equation can be evaluated for the two halfspaces
- It can be seen that

$$X^T n + b = 0 \forall X \in \text{line}$$

$$X^T n + b > 0 \forall X \in \text{subspace in the } n \text{ direction } (X_3)$$

$$X^T n + b < 0 \forall X \in \text{subspace in the } -n \text{ direction } (X_2)$$

Linear Algebra

Now that we have described what a hyper plane is, let me move on to the concept of half space to explain the concept of half space. I am going to look at this 2 dimensional picture on the left hand side of the screen. So, here we have a 2 dimensional space in X_1 and X_2 and as we have discussed before an equation in two dimensions would be a line which would be a hyperplane. So, the equation to the line is written as $X^T n + b = 0$. So, for, in this two dimensions we could write this line as; for example, $X_1 n_1 + X_2 n_2 + b = 0$, while I have drawn this line only for part of this picture. In reality this line would extend all the way on both sides.

Now, you notice the following. You see when I extend this line all the way on both sides, then this whole two dimensional space is broken into two spaces, one on this side of a line and the other one on this side of a line. Now these two spaces are what are called the half spaces. Now the question that we have is the following.

If there are points on one half space and points on the other half space, is there some characteristic that separates them? For example, can I do some computations for all the points on one half space and get some value and some computation for all the points on the other half space and get some value and use that to make some decisions and that is a reason why we are interested in this half spaces from a data science viewpoint.

So, this question is of importance in a particular kind of problem called a classification problem. Let me explain what that means. In fact, we are going to look at a very specific classification problem called binary classification problem. So, let us assume that I have, let us say in two dimensional space a data belonging to two classes.

For example, let us say I have data belonging to class one like this, and I call it class one and then I have data belonging to class two is something like this, call it class 2. So, these classes could be anything. So, for example, this could be a group of people who like South Indian restaurants and this could be a group of people, who do not like South Indian restaurants, and the coordinates X_1 and X_2 could be some way of characterizing people in terms of some attributes of these folks. Let us say we have taken a survey to say whether they like South Indian food or do not like South Indian food.

Now what we want to do is, if I give you the attributes of a new person, let us say that attribute falls here and then I ask you this question as to would this person like South Indian food or not like South Indian food and the answer would most likely be that this person will not like South Indian food, because this data point is very close to class 2.

Whereas if I gave you another point here for example, then you would come to the conclusion, this person is likely to like South Indian food. So, what we want to do is, we want to be able to evaluate cases like this. So, we want to somehow come up with a discriminating function between these two classes. So, one way to do that would be something like this; draw a line between these two classes and then say, if there is some characteristic that holds for this side of the line, which is what we called as a half space here. And if there is some characteristic that holds to this side of the line then we could use that characteristic as a discriminant function for doing this binary classification problem. So, that is the data science interest in understanding this topic in linear algebra.

Now let us proceed to see how we do this through some simple geometric concept. Let us go back to this picture and then ask the question as to how do I determine which side of a half plane or a half space, which half space does a point lie in. So, to understand this, what we are going to do is, we are going to take three points as shown here X_2 , X_1 and X_3 and ask the question as to how do I distinguish whether the point is on a line or to one half space or the other. So, the way we are going to do this, is the following. We are going to first look at this in little more detail and we know that when I write an equation of the form $X^T n + b = 0$, n is normal to this line, is something that we have already described.

However there is an important point to note. Here the normal could be defined in two ways, one is the normal is in this direction, the other thing to do is to just take the opposite direction and then define a normal in this fashion also. So, its important to know in which side normal is defined to understand this. For example, if I say this is a normal for an equation which is $X^T n + b = 0$.

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Data science for Engineers

Halfspace

- We can observe that the equation can be evaluated for the two halfspaces
- It can be seen that

$$X^T n + b = 0 \forall X \in \text{line}$$

$$X^T n + b > 0 \forall X \in \text{subspace in the } n \text{ direction } (X_3)$$

$$X^T n + b < 0 \forall X \in \text{subspace in the } -n \text{ direction } (X_2)$$

Linear Algebra

If I simply multiply this equation by - 1, then I am defining a normal to the other side. So, this is an important point to remember. Now what we want to know is, where do this points X_1 , X_2 , X_3 lie to do this. What we are going to do is, we are going to evaluate a discriminant function or a function which is basically the equation of the line. So, what we want to do is.

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Data science for Engineers

Halfspace

- We can observe that the equation can be evaluated for the two halfspaces
- It can be seen that

$$X^T n + b = 0 \forall X \in \text{line}$$

$$X^T n + b > 0 \forall X \in \text{subspace in the } n \text{ direction } (X_3)$$

$$X^T n + b < 0 \forall X \in \text{subspace in the } -n \text{ direction } (X_2)$$

Linear Algebra

We want to understand what this will be, what this will be and what this will be. Now, when we look at point X_1 we know that the point lies on the line. So, this is going to be 0. So, this is straightforward. What we are interested in, is what happens to this quantity for X_3 and X_2 , and is there some way in which we can say that every point to one side of the line will have the same characteristic and every other point on the other side of the line will have a different characteristic. So, to do this, let us first look at $X_3^T n + b$ and then see what happens.

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Data science for Engineers

Halfspace

- We can observe that the equation can be evaluated for the two halfspaces
- It can be seen that

$$X^T n + b = 0 \forall X \in \text{line}$$

$$X^T n + b > 0 \forall X \in \text{subspace in the } n \text{ direction } (X_3)$$

$$X^T n + b < 0 \forall X \in \text{subspace in the } -n \text{ direction } (X_2)$$

Linear Algebra

So, I want to know what this is. Notice in this picture I have defined a new point X on the line and then I have another vector which goes from X' to X_3 . Now X_3 is the vector that goes from here to here. From vector addition we know that I can write X_3 as X' , this + this $X' + Y'$. So, what I am going to do is, I am going to simply substitute this into

the equation and then see what happens. So, I am going to have $X'^T n + b$.

This is what I want to evaluate. This will become $X'^T n + b + Y'^T n$. All I have done is, I moved b closer to this term to show you something. Now notice what happens to this term right here, since X' is on the line and the equation of line is $X'^T n + b$ this has to go to 0. So, when we compute $X_3'^T n + b$, we are simply left with this term right here. And if you notice this term you would see that this is a dot product between this vector and this vector, and the most important thing to note here is the following, as long as the point lies to this side, this side of the line then you would see whatever point you take, the angle between that point and the normal would be in the following ranges.

So, you take any point this side or this side. So, the angle between the normal and that point is going to be the following. So, supposing we look at this and then say; I am going to do this angle in this direction right. So, what you are going to notice is the following. If the point is between these two, then I am going to have a positive θ angle. Now the way you do this is the following.

So, you go like this. So, for this quadrant if you start with 0 here for this quadrant the angle is going to be between 0 and 90, and for this quadrant the angle is going to be between 270 and 360. So, if a point is this side, the angle between this vector and this normal is going to be between 0 and 90. And if the point is in this side the angle is going to be between 270 and 360. We also know that when I have dot products $A^T b$, I can also write this as magnitude of a magnitude of $b \cos \theta$, where θ is the angle between these two vectors.

So, we will look at all the points up to here. So, whatever is a point you have these angles and all of these angles are between 0 and 90. So, for any point between here and here in this whole space you are going to get a b , some angle between 0 and 90, and we know from our high school rule, all silver teacups $\cos \theta$ will always be positive. So, $A^T b$ is going to be positive; that means, this is going to be positive. Now when you get two points here then the angles are going to be between 270 and 360 which is in the fourth quadrant. Again using the same rule all silver teacups the fourth quadrant is $c \cos$. So, $\cos$ is going to be positive. So, again you have $A^T b$ being positive.

So, irrespective of where the point is to this side of the line, when I take this $X_3'^T n + b$, I am always going to get a positive value. Now by similar argument you can say for any point on the other side or the other half space, the angles are going to be between 90 to 180 here and 180 to 270 and as we know $\cos \theta$ for angles between 90 to 270 is negative.

So, any point on this side of the line or the half space the computation $X^T n + b$ is going to be less than 0. So, this is an important idea that that I would like you to understand. So, what this basically says is the following. If you were to simply take any point that I give you and then I evaluate $X^T n + b$, if that point is on the side of the normal half space then $X^T n + b$ will be positive, and if its on the half space in the opposite side then its going to be negative. And I already told you how this is important from a data science viewpoint.

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Data science for Engineers

Hyperplanes and halfspaces: Example

- Let us consider a 2D geometry with $n = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and $b = 4$

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$X^T n + b = 0$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 4 = 0$$

$$x_1 + 3x_2 + 4 = 0$$

- The hyperplane is the equation of a line
- The halfspaces corresponding to this hyperplane are

$$x_1 + 3x_2 + 4 > 0 : \text{Positive halfspace}$$

$$x_1 + 3x_2 + 4 < 0 : \text{Negative halfspace}$$

Handwritten notes on the slide:

- Points: $(-1, -1)$, $(1, -1)$, $(1, -2)$
- Calculations:
 - $x_1 + 3x_2 + 4 = 0$ for $(-1, -1)$
 - $1 - 3 + 4 = 2 > 0$ for $(1, -1)$
 - $1 - 6 + 4 = -1 < 0$ for $(1, -2)$

Linear Algebra

So, let us consider simple 2D geometry and then let us take n as $\begin{bmatrix} 1 \\ 3 \end{bmatrix}$ and b as 4. So, this would give me this equation for $X^T n + b$. Now let us say I take a point on three points for example. So, let me consider $-1 - 1$ as one point, let us also consider $1 - 1$ as another point and let us consider $1 - 2$ as another point and then see what happens. So, when I take the point $-1 - 1$ and I substitute into this $x_1 + 3x_2 + 4$. So, it will be $-1 - 3 + 4$. So, the point $-1 - 1$ will lead to $-1 - 1$. Sorry will lead to 0. So; that means, the point $-1 - 1$ is on the line, when I take the point $1 - 1$. So, this is going to be $1 - 3 + 4$. So, this is going to be 2, so positive.

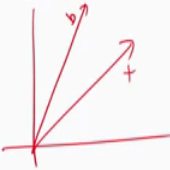
So, this is on in the positive half space and when I take the point $1 - 2$ then I am going to get $1 - 6 + 4$ which is going to be -1 less than 0. So, this is in the negative half space. So, this is on the hyperplane of the line, this is on the positive half space and this is on the negative half space. So, that tells you how to look at different points and then decide which side of the hyperplane or which half space these points lie.

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Data science for Engineers

Eigenvalues and eigenvectors

- We have previously seen linear equations of the form $Ax = b$
- What is the geometrical interpretation of this equation?
- We can make an interpretation as follows
 - When vector x is operated on by A , we obtain a new vector b with a different orientation





Linear Algebra

Now, that we have understood hyper planes and half spaces we are going to move on to the last linear algebra concept that I am going to teach in this module on linear algebra for data sciences. And once we are done with this topic, then we have enough information for us to teach you the various algorithms, commonly used algorithms or the first level algorithms in data science. So, let us look at this idea of eigenvalues and eigenvectors, we have previously seen linear equations of the form $Ax = b$. We have looked at it both algebraically and geometrically, we have spent quite a bit of time on looking at these equations algebraically. We talked about when these equations are solvable when there will be infinite number of solutions, how do we address all of those cases in a unified fashion and so on.

Now what we are going to do is, now that we know both vectors and so on, we are going to look at a slightly geometrical interpretation for this equation again and then explain the idea of eigenvalues and eigenvectors and then connect the notion of these eigenvalues and eigenvectors with the column space, null space and so on that we have seen before. So, this is very important, because these ideas are used quite a bit in data compression, noise removal, model building and so on. We will start saying I have this $Ax = b$ and A is an n by n matrix x is n by 1 and b is n by 1 . So, this is the kind of system that we are looking at.

So, we are going only look at square matrices n by n . Now you can think of this as n equations and n variables. There is also another interpretation you can give for this which is the following, supposing I have a vector x something like this and if I operate A on this. So, by operating, I mean we define an operation as pre multiplying this vector by A . So, let us say I operate A on this vector which is Ax then I notice

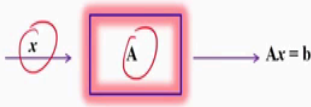
from this equation I get b , which is basically some other new direction that I have. So, you can think about this as the following I can think about this as a equation, which tells me that when I operate A on x then I get a new vector b which is in a different direction from x . So, this is a very simple interpretation of this equation $Ax = b$, which is what is written here x , I send it through a and I define sending it through a as pre multiplying by A ; so A times x equal b .

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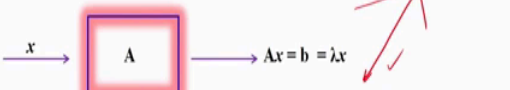
Data science for Engineers

Eigenvalues and eigenvectors

- Operator representation



- The newly obtained b vector represents a new orientation. So we ask the following question
- Are there directions for a matrix A such that when the matrix operates on these directions they maintain their orientation save for multiplication by a scalar (positive or negative)?
- That is



Linear Algebra 6

Now, that we have this interpretation, we ask the following question for a matrix A . Are there some directions which when you operate this A on they do not change their orientation. In other words I want to know if there are x vectors for matrix A such that when I operate A on x I get λx not b , λx here, the idea is because this is x , there is no change in orientation save multiplication by a scalar. Now this multiplication scalar could be positive or negative, in which case we are talking about the following.

So, if this is x . So, when I operate A on x , since its in the same direction, its

either this way or this way and if λ is positive, it will be in this direction and if λ is negative, it will be in this direction and so on. So, the question is, would there be directions like this for all kinds of matrices is an interesting question, that you could ask for.

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Data science for Engineers

Eigenvalues and eigenvectors

- The mathematical formulation of our question is

$$Ax = \lambda x$$
- The constant λ (*positive*) represents the amount of stretch or shrinkage the attributes x go through in the x direction
- The solutions (x) are known as eigenvectors and their corresponding λ are eigenvalues



Linear Algebra



Now let us focus on λ being positive, λ can be negative also. If λ is positive, then we see this equation and then notice that if λ is less than 1, then basically when I operate A on x the vector actually shrinks. So, if this is x , if λ is less than 1, this will be shrunk like this, and if λ is greater than 1 it will be at a higher magnitude than the original x vector.

Now the question is, for every matrix A , would there be A vectors like this x and what would be the scalar multiple and what is the use of all of this, is something that we should also address at some point as we go through this lecture. Now let me give you some definitions, this x are called eigenvectors and lambdas are called the eigenvalues corresponding to those eigenvectors. So, the questions that we are left with, are how do we find out that every matrix, whether it would have eigenvectors and how do I compute this eigenvectors and eigenvalues.

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Data science for Engineers

Eigenvalues and eigenvectors

- We can find the eigenvalues as follows

$$Ax = \lambda x \quad A(n \times n); x(n \times 1)$$

$$Ax - \lambda Ix = 0$$

$$(A - \lambda I)x = 0$$
- Thus the eigenvalues of the equation can be identified using

$$|A - \lambda I| = 0$$
- Substituting the eigenvalues in the original equation will help us find solutions for the eigenvector x

x = 0 (trivial solution)
x is in the nullspace of (A - λI)
(rank) + nullity = n

Linear Algebra

So, to compute the eigenvalues we follow this procedure that I am going to outline now. So, the original equation is $Ax = \lambda x$, what I could do is, I could bring this λx to this side, and then I get this equation $Ax - \lambda Ix = 0$. So, this becomes $A - \lambda I$ times $x = 0$. Now, notice that this is basically a vector equation, because I have n by n vector this is n by 1 . So, I have on the left hand side n by 1 .

So, I have a vector here and I want 0 s here. So, I want to find an x ; such that this is true. Now we have everything that we need to solve at this equation. So, what I am going to explain to you is the following. If I want to get an x which is not all 0 , notice that if x is all 0 , this is a solution right. So, $x = 0$ is a solution, but we are not interested in this solution, because this is what we call as a trivial solution.

We are not interested in this, we are only interested in solutions that we call as non trivial, at least one of the x s will have to be non 0 . Now notice that if this equation is solvable, then x is in the null space of $A - \lambda I$ matrix. This is something that we have seen before, while we define the null space and we also know that the rank nullity theorem says the rank of the matrix + nullity = n which is the number of columns, we are looking at square matrices n by n matrices. Now we know that if there is even one vector x ; such that this is 0 ; that means, then rank of the null space is at least 1 , and since the rank of the null space is at least one nullity is at least 1 ; that means, the rank of the matrix has to be less than n right, it can't be n , if this is n nullity is 0 , that means, there are no non trivial solutions.

So, if there needs to be a solution for x , then we know that the rank of the matrix $A - \lambda I$ has to be less than n ; that is the matrix $A - \lambda I$ is not a full rank matrix, and we know that if the matrix is not full rank then the determinant of that matrix has to be 0. So, in summary if we want a non trivial solution for x , then that necessarily means that this determinant $A - \lambda I$ has to be $= 0$.

Now once we solve for this equation and compute $A \lambda$, then we can go back and then substitute the value of λ here and then we have $A - \lambda I$ times $x = 0$, the way we have chosen λ is such that this matrix does not have full rank; that means, there is at least one vector in the null space, and using concepts that we have learned before we can identify this null space vector which would become the eigenvector.

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Data science for Engineers

Eigenvalues and eigenvectors: Examples

- Consider the following example with the given A matrix

$$A = \begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} \lambda x_1 \\ \lambda x_2 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 8 & 7 \\ 2 & 3 \end{vmatrix} - \lambda \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = \begin{vmatrix} 8-\lambda & 7 \\ 2 & 3-\lambda \end{vmatrix}$$

$$= 0$$

$$(8-\lambda)(3-\lambda) - 14 = 0$$

$$\lambda^2 - 11\lambda + 10 = 0 \quad \checkmark$$

$$\lambda = (10, 1)$$

Handwritten notes: $\lambda_1 = 10$, $\lambda_2 = 1$

- Thus we identify two eigenvalues and now we proceed to find the corresponding eigenvectors

R Code

```
A = matrix(c(8,7,2,3), 2, 2, byrow=TRUE)
ev = eigen(A)
values = ev$values
```

Console output

```
> values
[1] 10 1
```

Linear Algebra



Let me illustrate this with an example here, let us consider the matrix A which is $8 \ 7 \ 2 \ 3$ and let us compute this determinant $A - \lambda I$. So, you get the following equation and you get a quadratic equation here. Notice an interesting thing here; if I have an n by n matrix, the determinant in λ would be an n th order polynomial. In this case I have A_2 by 2 matrix. So, the determinant is a λ function which is a quadratic, and if its 3 by 3 it will be cubic and so on. So, this opens out the possibility of a solution to this equation being complex also. So, this is an important point to note here though your original matrix A is real. The solution to your eigenvalue problem could be either real or complex, depending on the polynomial that you end up with. In this case we have chosen this example in such a manner that I get two real solutions and the real solutions are 10 and 1. So, I can easily see that this equation has solutions 10 and 1. So; that means, I have two

eigenvalues $\lambda_1 = 10$ and $\lambda_2 = 1$. Now how do I go ahead and calculate the eigen vectors corresponding to these eigen values.

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Data science for Engineers

Eigenvalues and eigenvectors: Examples

• $\lambda = 1$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8x_1 + 7x_2 \\ 2x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$x_1 + x_2 = 0$$

Thus the eigenvector (unit) corresponding to $\lambda = 1$ is

$$X = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Handwritten notes:

- $8x_1 + 7x_2 = x_1 \Rightarrow 7x_1 + 7x_2 = 0 \Rightarrow x_1 + x_2 = 0 \checkmark$
- $2x_1 + 3x_2 = x_2 \Rightarrow 2x_1 + 2x_2 = 0 \Rightarrow x_1 + x_2 = 0 \checkmark$
- $\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2} = \sqrt{\frac{1}{2} + \frac{1}{2}} = \sqrt{1} = 1$

Linear Algebra 10

So, let us illustrate this for $\lambda = 1$. So, I take this eigenvalue eigen-vector equation now that I know $\lambda = 1$, this becomes $\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Now this turns out into these two equations, and if you notice you take the first equation, the first equation is $8x_1 + 7x_2 = x_1$.

So, if I take x_1 to this side I get $7x_1 + 7x_2 = 0$, which is the same as $x_1 + x_2 = 0$. If you take the second equation you will see that it is $2x_1 + 3x_2 = x_2$ which basically says $2x_1 + 2x_2 = 0$, which also is $x_1 + x_2 = 0$. So, both these equations turn out to be the same. Now any solution where x_2 is the negative of x_1 would be a eigenvector, what we do is, the following of all of those solutions.

We also make sure that we get an eigenvector which has unit magnitude. So, if you notice here the eigenvector that we get, you notice that x_1 , and this is x_2 and you notice that x_2 is $-x_1$ or x_1 is $-x_2$, which is what will satisfy this equation and instead of picking any k - k as a solution here, we pick a k in such a way that the magnitude of this vector is 1. So, we know that the magnitude of this vector will be 1 by root 2 whole square + - 1 by root 2 whole square root which will be root of half + half which will be root of 1 = 1. So, that way we make this a specific eigenvector which is unit length. We could do the same thing for λ equal 10, by much the same procedure you will notice that you will get this equation here $7x_2 = 2x_1$.

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Data science for Engineers

Eigenvalues and eigenvectors: Examples

- $\lambda = 10$

$$\begin{bmatrix} 8 & 7 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 10x_1 \\ 10x_2 \end{bmatrix}$$

$$\begin{bmatrix} 8x_1 + 7x_2 \\ 2x_1 + 3x_2 \end{bmatrix} = \begin{bmatrix} 10x_1 \\ 10x_2 \end{bmatrix}$$

$$7x_2 = 2x_1 \quad \checkmark$$
- Thus the eigenvector (unit) corresponding to $\lambda = 10$

$$X = \begin{bmatrix} 7 \\ \sqrt{53} \\ 2 \\ \sqrt{53} \end{bmatrix}$$

R Code

```
A = matrix(c(8,7,2,3), 2, 2, byrow=TRUE)
ev = eigen(A)
vectors <- ev$vectors
> vectors
[,1] [,2]
[1,] 0.9615239 -0.7071068
[2,] 0.2747211 0.7071068
```

Linear Algebra



So, basically what you could do is, any vector which is such that if X_1 is kX_2 is $2k$ by 7 would satisfy this equation; however, what we do is, we choose this k in such a way that the magnitude of the eigenvector is 1. So, in this case 7 by root 53 2 by root 53 , if you do the magnitude of this you will see this is going to be root of 49 by $53 + 4$ by 53 , which will be root of $1 = 1$. So, you see that the magnitude is 1 and also this equation is basically satisfied by any eigenvector which is of this form k to k by 7 .

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Data science for Engineers

Summary

- $Ax = b$ • Geometric interpretation
- $Ax = \lambda x$ • Eigenvalue-eigenvector equation
- λ • N eigenvalues from $|A - \lambda I| = 0$
- x • Eigenvectors, generally expressed in unit vector form

Linear Algebra

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So, in summary for the eigenvalue eigenvector portion of this lecture, we started with $Ax = b$ which has a geometric interpretation of

A operating on x giving a new vector b . Now if we force this b to be λx some scalar multiple of x itself, where the scalar multiple could be either positive or negative, we get the eigenvalue eigenvector equation and to calculate the eigenvalue. What we do is, we calculate the determinant $A - \lambda I$. I set it to 0 for an n by n matrix, there will be an n th order polynomial that we need to solve which opens out to the possibility of the eigenvalues, being either real or complex. And once we identify the eigenvalues we can get eigenvectors as the null space of $A - \lambda I$ where λ is the corresponding eigenvalue.

In the next lecture what I will do is, I will connect this notion of eigenvalues and eigenvectors to things that we have already talked before in terms of column space and null space of matrices and so on. We already saw that the eigenvectors are actually in the null space of $A - \lambda I$, I am going to develop on this idea and then show you other connections between eigenvectors and these fundamental subspaces, and I will also allude to how this is a very important problem; that is used in a number of data science algorithms. So, I will see you in the next class.

Thank You.

Data Science for Engineers
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Department of Computer Science and Engineering
Indian Institute of Technology, Madras

Lecture - 18
Linear Algebra - Distance, Hyperplanes and Halfspaces, Eigenvalues, Eigenvectors

This is the last lecture in the series of lectures on Linear Algebra for data science and as I mentioned in the last class, today, I am going to talk to you about the connections between eigenvectors and the fundamental subspaces that we have described earlier. We saw in the last lecture that the eigenvalue eigenvector equation results in

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Connections between eigenvectors, column space and null space

- We know that eigenvalues can be complex numbers even for real matrices
- When eigenvalues become complex, eigenvectors also become complex
- However, if the matrix is symmetric, then the eigenvalues are always real
- As a result, eigenvectors of symmetric matrices are also real
- Further, there will always be n linearly independent eigenvectors for symmetric matrices

$$|A - \lambda I| = 0$$

$$P_A(\lambda) = 0$$

$$Ax = \lambda x$$

$$A = A^T$$

$$A = \begin{pmatrix} \lambda_1 & \lambda_2 & \dots & \lambda_n \\ \theta_1 & \theta_2 & \dots & \theta_n \\ \vdots & \vdots & \ddots & \vdots \\ \theta_n & \theta_{n-1} & \dots & \theta_1 \end{pmatrix}$$

$$P_A(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2) \dots (\lambda - \lambda_n)$$

$$\lambda = 1 \Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

this equation having to be satisfied which is $A - \lambda I = 0$. In general, we also saw that, this would turn out to be a polynomial of degree n in λ , which basically means that even if this matrix A is real, because the solutions to a polynomial equation could be either real or complex, you could have eigenvalues that are complex.

So, for a general matrix, you could have eigenvalues which are either real or complex. And notice that since we write the equation Ax

$= \lambda x$, whenever this eigenvalues become complex, then the eigenvectors are also complex vectors. So, this is true in general; however, if the matrix is symmetric and symmetric matrices are of the form $A = A^T$, then there are certain nice properties for these matrices which are very useful for us in data science. We also encounter symmetric matrices, quite a bit in data science for example, the covariance matrix turns out to be a symmetric matrix and there are several other cases where we deal with symmetric matrices.

So, these properties of symmetric matrices are very useful for us when we look at algorithms in data science. Now, the first property of symmetric matrices that is very useful to us is; if the matrix is symmetric, then the eigenvalues are always real. So, irrespective of what that symmetric matrix is, this polynomial would always give real solutions for symmetric matrices. And as I mentioned before if this turns out to be real, then the eigenvectors are also real. Now, there is another aspect of an eigenvalues and eigenvectors that is important; if I have a matrix A and I have n different eigenvalues λ_1 to λ_n , all of them are distinct, then I will definitely have n linearly independent eigenvectors corresponding to them which could be v_1 ; v_2 all the way up to v_n ; however, if there are certain eigenvalues which are repeated. So, for example, if we take a case where eigenvalue λ_1 is repeated, then I could have some polynomial, which is like this.

So, the polynomial, original polynomial has eigenvalue; λ_1 repeated twice and then there is another $n - 2$ order polynomial, which will give you $n - 2$ other solutions. Now, in this case, when I have λ_1 repeated like this, then it could turn out that this eigenvalue either has 2 eigenvectors, which are independent or it might have just one eigenvector.

So, finding n linearly independent eigenvectors is not always guaranteed for

any general matrix and we already know that eigenvectors could be complex for any general matrix; however, when we talk about symmetric matrices, we can say for sure that the eigenvalues would be real, the eigenvectors would be real, further we are always guaranteed that we will have n linearly independent eigenvectors for symmetric matrices. It does not matter how many times the eigenvalues get repeated. One classic example of a symmetric matrix, where eigenvalues are repeated many times, so take identity matrix, something like this here, this identity matrix has eigenvalue $\lambda = 1$, which is repeated thrice.

But, it would have three independent eigenvectors; $1, 0, 0$; $0, 1, 0$ and $0, 0, 1$. So, this is a case where eigenvalues repeated is thrice, but there are three independent eigenvectors. So, this is also an important result that we should keep in mind.

(Refer Slide Time: 05:04)

Data science for Engineers

Connections between eigenvectors, column space and null space

- Symmetric matrices have a very important role in data sciences
- In fact symmetric matrices of the form $A^T A$ or AA^T are often encountered
- Eigenvalues of matrices of the form $A^T A$ or AA^T while being real are also non-negative
- As discussed for general symmetric matrices, there will be n linearly independent eigenvectors for matrices of this form also
- What is the connection between the eigenvectors and the column space and null space of a (symmetric) matrix ?

$$\begin{aligned} (A^T A)^T &= A^T (A^T)^T \\ &= A^T A \\ &= A^T A \end{aligned}$$



1. Linear Algebra

And as I mentioned in the last slide, Symmetric matrices have a very important role in data sciences. In fact, symmetric matrices of the type, $A^T A$ or AA^T are often encountered in data science computations. And notice that both of these matrices are symmetric. So, for example, if I take $A^T A$, this will be $A^T A$; $A^T A$, which will be $A^T A$. So, the transpose of the matrix is the same. You can verify that, AA^T transpose is also symmetric through the same idea. So, we know matrices of the form $A^T A$ or AA^T are both symmetric and they are often encountered; when we do computations in data science. And we know from the previous slide, I had mentioned that the eigenvalues of symmetric matrices are real, if the symmetric matrix also takes this form or this form.

We can also say that while the eigenvalues are real; they are also non-negative, that is they will be either 0 or positive, but none of the eigenvalues will be negative. So, this is another important idea that we will use; when we do data science, when we look at covariance matrices and so on. Also the fact that, this $A^T A$ and A^T are symmetric matrices; guarantees that there will be n linearly independent eigenvectors for matrices of this form also. So, what we are going to do right now is, because of the importance of symmetric matrices in data science computations, we are going to look at the connection between the eigenvectors and the column space a null space for a symmetric matrix. Some of these results translate to non-symmetric matrices also, but for symmetric matrices, all of these are results that we can use.

(Refer Slide Time: 07:04)

Data science for Engineers

Connections between eigenvectors, column space and null space

$Av = \lambda v$

- What happens when the eigenvalues become zero?
 $Av = 0$ $\Rightarrow \lambda = 0$ is a nullspace vector $AB=0$ ✓
- The eigenvectors corresponding to zero eigenvalues are in the null space of the matrix
- Conversely, if the eigenvalue corresponding to an eigenvector is not zero then that eigenvector cannot be in the null space
 $Av=0$ and not null vector ✓ if A is full rank

Linear Algebra 4

So, we go back to the eigenvalue eigenvector equation; Av is λv . And this result that we are going to talk about right now, is true whether the matrix is A symmetric or not. If $Av = \lambda v$, we ask the question, what happens when λ is 0? That is one of the eigenvalues becomes 0. So, when one of the eigenvalues becomes 0, then we have this equation which is $Av = 0$. So, we can interpret v as an eigenvector corresponding to eigenvalue 0.

We have also seen this equation before, when we talked about different sub-spaces for matrices; we saw that null space vectors are of the form $A\beta = 0$ from one of our initial lectures. You notice that, this and this form are the same. So, that basically means that, v which is an eigenvector corresponding like corresponding to eigenvalue, $\lambda = 0$, is a null space vector, because it is just of the form that we have here. So, we could say, the eigenvectors corresponding to 0 eigenvalues are in the null space of the original matrix A . Conversely, if the eigenvalue corresponding to an eigenvector is not 0, then that eigenvector cannot be in the null space of A . So, these are important results that we need to know.

So, this is how eigenvectors are connected to null space. If none of the eigen-values are zero, that basically means that the matrix A is full rank and; that means, that I can never solve $Av = 0$; and get non trivial v . So, it is not possible, if A is full rank. So, if A is full rank, I cannot solve for this and get non trivial v . So, whenever λ is; λ is such that, there are there is no eigenvalue that is zero; that means, A is full rank matrix; that means, there is no eigenvector such that $Av = 0$ which basically means that there are no vectors in the null space.

(Refer Slide Time: 09:40)

Data science for Engineers

Connections between eigenvectors, column space and null space

- Let us assume that there are r eigenvectors corresponding to zero eigenvalue
- This means that the null space dimension is r
- From rank-nullity theorem (discussed before), we know that the column rank should be $n - r$
- That is $n - r$ independent vectors are enough to represent all the vectors in the columns of the matrix (column space)
- What could be a basis for this column space or what could be the $n - r$ independent vectors?

Handwritten notes:
 A is Symmetric
 r zero eigenvalues
 $n-r$ non-zero eigenvalues
 r eigenvectors
 $\text{rank} + \text{nullity} = n$
 $\text{rank} = n - r$

Linear Algebra



Now, let us see the connection between eigenvectors and column space. In this case, I am going to show you the result; and this result is valid for symmetric matrices. Let us assume that I have a symmetric matrix A ; and the symmetric matrix A , we know will have n real eigenvalues. Let us assume that r of these eigenvalues are 0.

So, this r could be 0 also; that means, there is no eigenvalue which is zero. So, even then all of this discussion is valid. But as a general case, let us assume that r eigenvalues are 0. So, there are r zero eigenvalues. And since we are assuming this matrix is n by n , there will be n real eigenvalues of which r are 0. So, there will be $n - r$ non-zero eigenvalues. And from the previous slide, we know that the r eigenvectors corresponding to this r 0 eigenvalues are all in the null space ok. So, since I have r 0 eigenvalues, I will have r eigenvectors corresponding to this.

So, all of these r eigenvectors are in the null space which basically means that the dimension of the null space is r ; because there are r vectors in the null space; and from rank-nullity theorem, we know that $\text{rank} + \text{nullity} = \text{number of columns in this case } n$; since there are r eigenvectors in the null space, nullity is r . So, the rank of the matrix has to be $= n - r$.

So, that is what we are saying here. And further we know that column rank = row rank; and since the rank of the matrix is $n - r$, the column rank also has to be $n - r$. This basically means that there are $n - r$ independent vectors in the columns of the matrix. So, one question that we might ask is the following; we could ask what could be a basis set for this column space? Or what could be the $n - r$ independent vectors that we can use as the columns subspace?

(Refer Slide Time: 12:24)

Data science for Engineers

Connections between eigenvectors, column space and null space

- Notice that there are $n - r$ eigenvectors which are not in the null space
- We know that these are independent
- We also know that these vectors are a linear combination of all the column vectors – that is they are in the column space

$$Av = \lambda v$$

- Further, we know that the dimension of the column space is $n - r$ (rank-nullity theorem)
- This implies that the eigenvectors corresponding to the non-zero eigenvalues form a basis for the column space

A is symmetric
 $\hookrightarrow$ n LI eigenvectors

$A \begin{bmatrix} v_1 \\ \vdots \\ v_{n-r} \end{bmatrix} = \begin{bmatrix} \lambda_1 v_1 \\ \vdots \\ \lambda_{n-r} v_{n-r} \end{bmatrix}$
 $v = \left(\frac{1}{\lambda}\right) A + \dots + \left(\frac{1}{\lambda}\right) A_n$
 $v = c_1 A_1 + \dots + c_n A_n$
 $A_1, \dots, A_n$

So, there are a few things that we can notice based on what we have discussed till now. First, notice that the $n - r$ eigenvectors that we

talked about in the last slide, the ones that are not eigenvector is corresponding to $\lambda = 0$; they cannot be in the null space; because λ is a number which is different from 0. So, these $n - r$ eigenvectors cannot be in the null space of the matrix A . So, let me write again. We are discussing all of this for symmetric matrices. We know, that all of this $n - r$ eigenvectors are also independent; because we said irrespective of what the symmetric matrix is, we will always get n linearly independent eigenvectors.

So, that means, these $n - r$ eigenvectors are also independent. We also know that each of these independent eigenvectors are going to be linear combinations of columns of A . To see this, let us look at this equation. So, let me write this out. So, I could call this as A , I am going to expand this v , into v_1, v_2 , all the way up to v_n ; notice that these are components of v . We are just taking one eigenvector v and then these are the n components in that eigenvector. I can write this as λv and from the previous lecture of how to do this column multiplication and how to interpret this column multiplication, I said you could think of this as v_1 times the first column of A + v_2 times the second column of A ; all the way up to v_n times the n th column of A = λv . Now in this equation, let me be very clear; these are scalars which are components in the eigenvector v ; these are column vectors; this is a first column of A , second column of A , this is n th column of A , this is again a scalar λ ; which is the eigenvalue corresponding to v .

So, this could be true for any of the $n - r$ eigenvectors; which are not in the null space of this matrix A . Now, take λ to the other side. So, you will have this equation as v , is v_1 by λA_1 and so on + v_n by λA_n . Again v_1 is a scalar λ is a scalar. So, these are all constants that we are using to multiply these columns. Now you will clearly see that, each of these eigenvectors; $n - r$ eigenvectors are linear combinations of the columns of A . So, there are $n - r$ linearly independent eigenvectors like this and each of these are combinations of columns of A . And we also know that the dimension of the column space is $n - r$. In other words, if you take all of these columns, A_1 to A_n ; these can be represented using just $n - r$ linearly independent vectors.

Now, when we put all of these facts together, which is the $n - r$ eigen-vectors are linearly independent; they are combinations of columns of A ; and the number of independent columns of A can be only $n - r$. So, this implies that the eigenvectors corresponding to the non-zero eigenvalues for a symmetric matrix form a basis for the column space. So, this is the important result that I wanted to show you, with all of these ideas. Now again these results we will see and use as we look at some of the data science algorithms later.

(Refer Slide Time: 16:20)

Data science for Engineers

Example

- Consider the following A matrix

$$A = \begin{bmatrix} 0.36 & 0.48 & 0 \\ 0.48 & 0.64 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

- Notice that this is a symmetric matrix
- The eigenvalues for this matrix are $\lambda = (0, 1, 2)$
- The eigenvectors corresponding to these eigenvalues are

$$v_1 = \begin{bmatrix} -0.8 \\ 0.6 \\ 0 \end{bmatrix}, v_2 = \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Handwritten notes on the slide:

- $A^T = A$
- $|A - \lambda I| = 0$
- $P_3(\lambda) = 0$
- $Ax = \lambda x$
- $3-1=2$
- $A^T = A$

Linear Algebra



So, let us take a simple example to understand how all of these work. Let us consider a matrix which is of this form here; it is a 3 by 3 matrix. First thing that I want you to notice, that this is a symmetric matrix. So, if you do $A^T = A$. And we said symmetric matrices will always have real eigenvalues and when you do the eigenvalue computation for this, the way you do the eigenvalue computation is, you take determinant $A - \lambda I = 0$, then you are going to get a third order polynomial, you set it = 0; and then you calculate the three solutions to this polynomial and these would turn out to be the solution 0, 1, 2 and you take each of these solutions and then substitute it back in and then solve for $Ax = \lambda x$.

Then you would get the three eigenvectors corresponding to this which are given by this, this and this. I have noticed from our discussion before; since this is an eigenvector corresponding to $\lambda = 0$; so, this is going to be in the null space of this matrix A and these are the remaining 2; how do I get this 2? Which is $3 - 1$, n, n by n. So, it is A_3 by 3 matrix and nullity is 1; because there is only one eigenvector corresponding to $\lambda = 0$. So, I get 2 other linearly independent vectors. And in the last slide, when we were discussing the connections, we claim that these two eigenvectors will be in the column space or in other words, what we are claiming is that these three columns can simply be written as a linear combination of these two columns; and we are also sure that when we do A times v_1 , this will go to 0. So, let us verify all of this in the next slide.

(Refer Slide Time: 18:26)

Data science for Engineers

Example

- From our prior understanding, the eigenvector corresponding to the zero eigenvalue will be in the null space
- We check that

$$Av_1 = \begin{bmatrix} 0.36 & 0.48 & 0 \\ 0.48 & 0.64 & 0 \\ 0 & 0 & 2 \end{bmatrix} \begin{bmatrix} -0.8 \\ 0.6 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

- Interestingly, in the initial lectures, it was identified that the null space vector identifies a relationship between the variables
- Hence, the eigenvector corresponding to the zero eigenvalue can be used to identify the relationships among variables

Eigenvectors to zero eigenvalue

Linear Algebra



So, let us first check A times v_1 . So, this is a matrix, I have a times v_1 here and you can quite easily see that when you do this computation, you will get this 0, 0, 0; which basically shows that this is the eigenvector corresponding to zero eigenvalue. Interestingly, in our initial lectures, we talked about null space and then we said the null space vector identifies a relationship between variables. Now, since this eigenvector is in the null space, the eigenvector corresponding to the eigenvector, corresponding to zero eigenvalue or eigenvectors corresponding to zero eigenvalues, identify the relationships between the variables because these eigenvectors are in the null space of the matrix.

So, it is an interesting connection that we can make. So, the eigenvectors corresponding to zero eigenvalue can be used to identify relationships among variables.

(Refer Slide Time: 19:33)

Data science for Engineers

Example

- Let us now check if the other two eigenvectors shown below span the column space

$$v_2 = \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix}, v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

- This is demonstrated as below

$$A \begin{bmatrix} 0.36 \\ 0.48 \\ 0 \end{bmatrix} = 6 * \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix}$$

$$A \begin{bmatrix} 0.48 \\ 0.64 \\ 0 \end{bmatrix} = 8 * \begin{bmatrix} 0.6 \\ 0.8 \\ 0 \end{bmatrix}$$

$$A \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix} = 2 * \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

A_1, A_2, A_3 are linearly independent

Linear Algebra



Now, let us do the last thing that we discuss. Let us now check the other two eigenvectors shown below. So, this is for the other two eigenvalues, span the column space. So, what I have done here is, I have taken each one of these columns from matrix A. So, this is column 1. So, this is A_1 , this is A_2 and this is A_3 . Column 1 is 6 times v_2 , column 2 is 8 times v_2 and column 3 is 2 times v_3 . So, we can say that A_1 , A_2 and A_3 are linear combinations of v_2 and v_3 .

(Refer Slide Time: 20:29)

Data science for Engineers

Summary

- $Ax = \lambda x$ • Symmetric matrices have real eigenvalues ✓
- $Ax = \lambda x$ • Symmetric matrices also have n linearly independent eigenvectors ✓
- $Ax = 0$ • Eigenvectors corresponding to zero eigenvalues span the null space (A)
- $Ax = \lambda x$ • Eigenvectors corresponding to non-zero eigenvalues span the column space for symmetric matrices (A)

Linear Algebra 10

So, v_2 and v_3 form a basis for this column space of matrix A. So, to summarize, we have $Ax = \lambda x$ and we largely focused on symmetric matrices in this lecture. So, we saw that, if we have symmetric matrices, they have real eigenvalues. We also saw that symmetric matrices have n linearly independent eigenvectors. We saw that the eigenvectors corresponding to zero eigenvalues span the null space of the matrix A and eigenvectors corresponding to nonzero eigenvalues span the column space of A for symmetric matrices that we described in this lecture. So, with this, we have described most of the important fundamental ideas from linear algebra that we will use quite a bit in the material that follows.

The linear algebra parts will be used in regression analysis, which you will see as part of this course. And many of these ideas are also useful in algorithms that do classification for example, we talked about half spaces and so on; and the notion of eigenvalues and eigenvectors are used pretty much in almost every data science algorithm, of

particular node is one algorithm which is called the principle component analysis which we will be discussing later in this course where these ideas of connections between null space, column space and so on are used quite heavily.

So, I hope that we have given you a reasonable understanding of some of the important concepts that you need to learn to understand some of the material that we are going to teach in this course and as I mentioned before, linear algebra is a vast topic. There are several ideas; how, how do these ideas translate, which ones of these are applicable or not applicable for non-symmetric matrices and so on. And from the previous lectures, how do we develop some of those concepts more can be found in many good linear algebra books; however, our aim here has been to really call out the most important concepts that we are going to use again and again in this first course on data science for engineers, more advanced topics in linear algebra will be covered when we teach the next course on machine learning where those concepts might be more useful in advanced machine learning techniques that we will teach. So, with this we close this series of lectures on Linear Algebra and the next set of lectures would be on the use of statistics in data science.

I thank you and I hope to see you back after you go through the module on statistics which will be taught by my colleague professor Shankar Narasimhan.

Optimality Conditions: Equality Constraints (I)



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$$\mathbf{x}^* = \arg \min_{\mathbf{x}} f(\mathbf{x})$$

subject to $h_i(\mathbf{x}) = 0 \quad (i = 1, 2, \dots, m)$

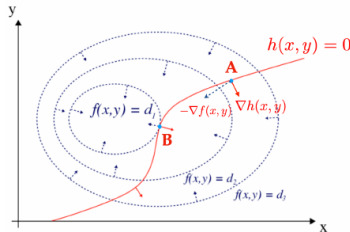
Theorem 1 (Lagrange necessary conditions)

Given $f(\mathbf{x})$ and $\{h_i(\mathbf{x})\}$ are differentiable. If a point $\mathbf{x}^*$ is a local optimum, then the gradients of these functions are linearly dependent at $\mathbf{x}^*$:

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i \nabla h_i(\mathbf{x}^*) = 0$$

where $\lambda_i \in \mathbb{R}$ are the Lagrange multipliers.

stationary points of $f(\mathbf{x})$ may not satisfy the constraints.



$$\min_{x,y} f(x,y)$$

subject to

$$h(x,y) = 0$$

Conditions: Equality Constraints (II)

- Theorem 1 suggests **the method of Lagrange multipliers**
- introduce a **Lagrange multiplier** $\lambda_i \in \mathbb{R}$ for each equality constraint
- construct the **Lagrangian** function:

Conditions: Equality Constraints (II)

- Theorem 1 suggests **the method of Lagrange multipliers**
- introduce a **Lagrange multiplier** $\lambda_i \in \mathbb{R}$ for each equality constraint
- construct the **Lagrangian** function:

$$L(\mathbf{x}, \{\lambda_i\}) = f(\mathbf{x}) + \sum_{i=1}^m \lambda_i h_i(\mathbf{x})$$

- minimize the **Lagrangian** function w.r.t. $\mathbf{x}$ and all Lagrange multipliers $\{\lambda_i\}$, converting a constrained optimization problem into an unconstrained one

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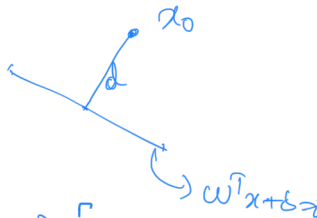
Example

distance from a point $\mathbf{x}_0 \in \mathbb{R}^n$ to a hyperplane $\mathbf{w}^T \mathbf{x} + b = 0$ in the space $\mathbf{x} \in \mathbb{R}^n$, where $\mathbf{w} \in \mathbb{R}^n$ and $b \in \mathbb{R}$ are given.

$$d^2 = \min_{\mathbf{x}} \|\mathbf{x} - \mathbf{x}_0\|^2$$

$$\mathbf{w}^T \mathbf{x} + b = 0 \quad \checkmark$$

$\mathbf{x} \in \mathbb{R}^n$



$$\mathcal{L}(\mathbf{x}, \lambda) = \|\mathbf{x} - \mathbf{x}_0\|^2 + \lambda (\mathbf{w}^T \mathbf{x} + b)$$

$$\mathcal{L}(\mathbf{x}, \lambda) = (\mathbf{x} - \mathbf{x}_0)^T (\mathbf{x} - \mathbf{x}_0) + \lambda (\mathbf{w}^T \mathbf{x} + b)$$

$$\frac{\partial \mathcal{L}(\mathbf{x}, \lambda)}{\partial \mathbf{x}} = 0 \Rightarrow 2(\mathbf{x} - \mathbf{x}_0) + \lambda \mathbf{w} = 0$$

$$\Rightarrow 2(\mathbf{x}^* - \mathbf{x}_0) + \lambda^* \mathbf{w} = 0$$



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$$\rightarrow x^* = x_0 - \frac{x^*}{2} w \quad \leftarrow (*)$$

$$w^T x^* + b = 0 \Rightarrow x^* = \frac{-(w^T x_0 + b)}{\|w\|^2} \quad \leftarrow (**)$$

$$d^2 = \|x^* - x_0\|^2 = \frac{(w^T x_0 + b)^2}{\|w\|^2}$$

$$d = \frac{|w^T x + b|}{\|w\|}$$

From projection of vector lecture, i tries to these dataset project on xy-plane

Three variables (pH, Temperature, and EC), each row in your data represents a point or a vector in a three-dimensional space (R^3). In this space, the variables act as the coordinates x,y,z coordinate.



Temperature (EC ($\mu\text{S}/\text{cm}$)

2	6.4	21.8	720
3	6.7	22.5	745
4	6.6	23.1	760
5	6.8	22.9	780
6	6.5	21.5	735
7			

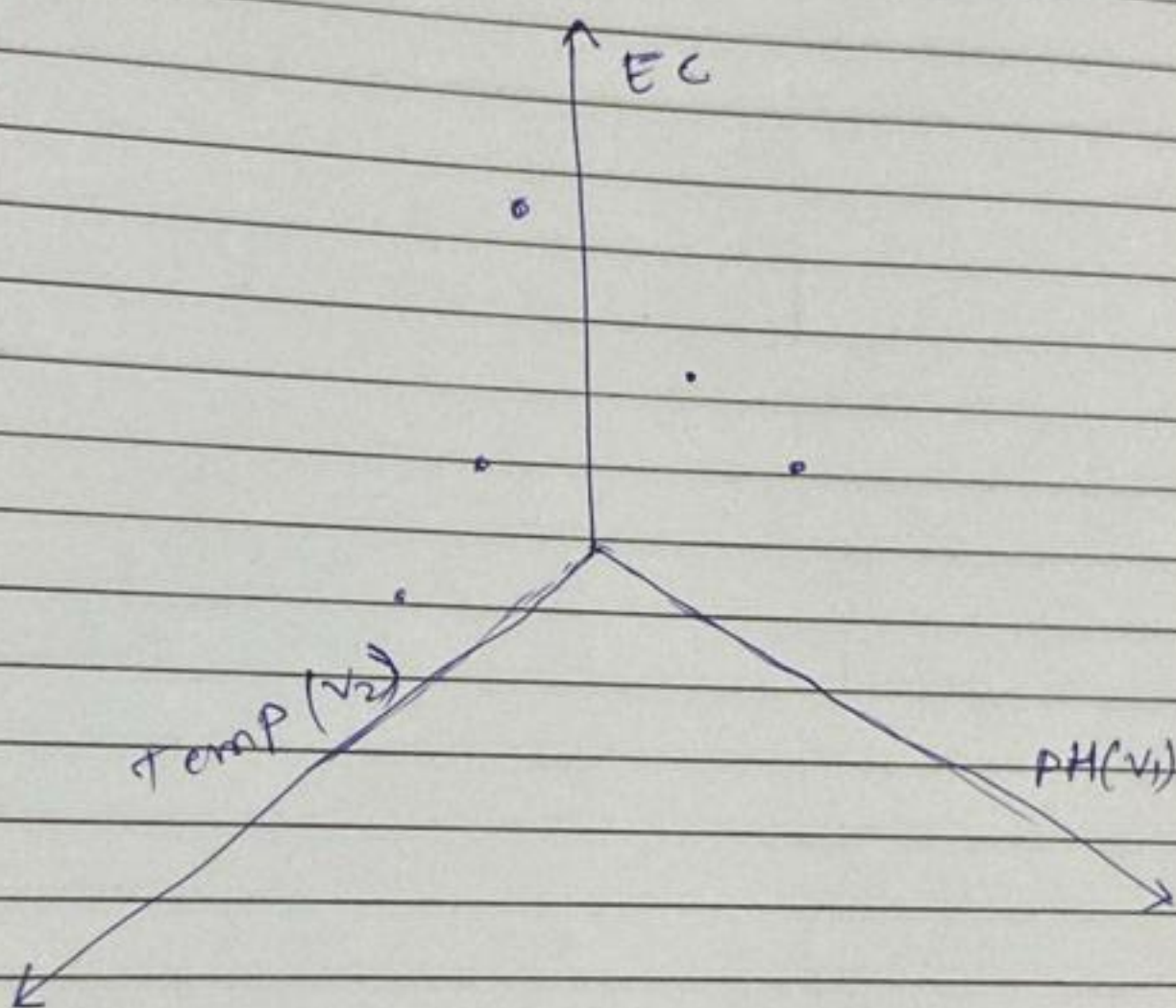
Each row of data is a vector X in $\mathbb{R}^3$ space.

$$\text{vector } X_1 = (6.4, 21.8, 720)^T$$

$$X_2 = (6.7, 22.5, 745)^T$$

$$X_4 = (6.8, 22.9, 780)^T, X_3 = (6.6, 23.1, 760)^T$$

$$X_5 = (6.5, 21.5, 735)^T$$



projection vector $\hat{X}_i$ in the lower dim of vector X_i into a lower dim is represent mathematically

$$\hat{X} = c_1 v_1 + c_2 v_2 \text{ \& vector addition}$$

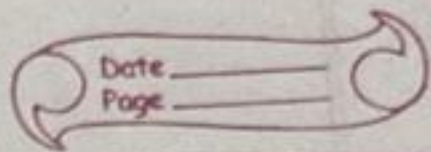
$$X = c_1 v_1 + c_2 v_2 + n \text{ (in } \mathbb{R}^3 \text{ space)}$$

$$v_1^T n = 0 \Rightarrow v_1^T (X - c_1 v_1 - c_2 v_2) = 0$$

$$\Rightarrow v_1^T X - c_1 v_1^T v_1 + c_2 v_1^T v_2 = 0$$

$$\Rightarrow \cancel{v_1^T X} = c_1$$

$$\hat{x} = \left(\frac{v_1^T}{v_1^T v_1} \right) v_1 + \left(\frac{v_2^T}{v_2^T v_2} \right) v_2$$



and in general, $\hat{x} = V(V^T V)^{-1} V^T x$, where
 $V = (v_1, v_2, \dots, v_n)$.

$x_i \in \mathbb{R}^3$ and $\hat{x}_i$ is orthogonal projection and
 V is ~~two~~ l.i. vector